# Tariffs, Rules of Origin, and the Reorganization of North American Supply Chains*

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**Abstract**

This paper quantifies how tariffs and rules of origin drive the reorganization of North American supply chains around Mexico, and what that reorganization implies for welfare. Between 2018 and 2025 three shocks reshaped this landscape: the U.S.-China trade war, the 2020 NAFTA-to-USMCA transition, and a 2025 U.S. executive order sharply raising tariffs on Mexican imports outside USMCA preference. I develop a global sourcing model in which heterogeneous firms choose source countries and, product by product, whether to export under USMCA, thus having to comply with rules of origin, and calibrate it on matched census-customs microdata. Through the lens of the model, USMCA protects Mexico from U.S. tariffs, given that firms respond to a higher tariff on non-compliant goods by complying with the rules of origin rather than paying it. The value of the agreement therefore grows sharply with that tariff, meaning that eliminating it would impose a welfare loss about six times larger at the tariff level set by the 2025 order than at the tariffs in place before it. In contrast, the transition from NAFTA to USMCA was roughly welfare-neutral for Mexico, while the shift in relative sourcing efficiency across countries raised welfare.

*JEL codes:* F13, F14, F15, L23.

*Keywords:* rules of origin, USMCA, global sourcing, heterogeneous firms.

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# 1 Introduction

Over the last several years North American trade policy has become increasingly complex. For nearly three decades, trade in the region operated under the North American Free Trade Agreement (NAFTA), which guaranteed frictionless commerce among the United States, Canada and Mexico as long as a sufficient share of each product was sourced within the region. Firms in turn built their continental supply chains around this certainty. This is no longer the case. Since 2018, a bilateral tariff war between the United States and China, the replacement of NAFTA by the more restrictive United States-Mexico-Canada Agreement (USMCA), and a sequence of emergency U.S. tariffs on goods that fall short of USMCA's regional content requirements have substantially altered what it costs to produce in Mexico and export to the other two nations. Preferential access can no longer be taken as given, and has instead become a changing policy margin that firms have to actively choose, comply with or forgo at every shipment.

Mexican manufacturing exports are at the center of the economy that these shocks are reorganizing. These exports amount to roughly a third of Mexican GDP, of which about 80% is destined for the United States, and are built on a long tradition of foreign investment in export-oriented plants, which today are embedded in the IMMEX program (the modern successor to the 1960s maquiladora regime). On the output side, firms face a choice on every shipment: they can either comply with USMCA's rules of origin and ship tariff-free to the U.S., or forgo preference and pay the Most Favored Nation (MFN) tariff. Given that this choice shapes where firms source their inputs, where they sell and how much they earn, movement on the compliance margin ends up propagating through the broader economy.

The 2018 trade war raised product-specific tariffs on half of the U.S.–China trade universe. Rather than triggering reshoring, these tariffs diverted trade toward "bystander" economies, where the sectoral specialization that each country already had determined how much of this trade it was able to absorb (Fajgelbaum et al., 2024). Given its geographic proximity, deep supply-chain linkages and preferential access under NAFTA, Mexico was a natural candidate to absorb part of this trade. Between 2017 and 2025, U.S. imports from China fell by roughly half in real terms, whereas U.S. imports from Mexico grew 31%. At the same time, Mexico's own imports from China grew 32%, with intermediate and capital goods accounting for roughly 60% of that growth.[1] The scale of this restructuring can also be observed on the ground, as foreign-headquartered firms alone added roughly half a million manufacturing workers in Mexico over the same period (Utar, 2026). It is important to mention, however, that these volumes do not tell us what happened on the preferential

[1] UN Comtrade bilateral flows at HS-6, mapped to end-use categories using the UN BEC concordance.

margin. When USMCA replaced NAFTA in July 2020 with tighter rules of origin, the share of Mexico-to-U.S. trade qualifying for preference did not increase but actually fell, from 56% in 2017 to 48% in 2024. It was only after March 2025, when an executive order under the International Emergency Economic Powers Act (IEEPA) imposed an addtional 25% (additional to the existing MFN rate) tariff on Mexican imports while exempting USMCA-compliant goods, that this share jumped to about 85%. Firms had been rerouting trade through Mexico for years, but increasingly outside of the preferential regime, and it was not until the cost of staying outside of it became large enough that they formalized this trade under USMCA. While these tariffs were eventually struck down by the U.S. Supreme Court, the episode provides a glimpse at how firms react when not complying with a free trade agreement becomes prohibitively expensive. Following the ruling, the tariff on Mexican goods outside USMCA preference currently stands at 10%. It is worth pointing out that the low compliance before 2025 does not mean that the agreement mattered little to Mexico. Rather, its value remained latent while the tariff that it waives was small, and the jump in compliance after March 2025 is the firm-level counterpart of the increase in that value, which I quantify in Section 6.5.

The goal of this paper is to study the trade-off that firms face between cost-minimizing global sourcing and free-trade-agreement compliance, and to quantify its welfare implications. The relevance of this margin goes beyond what has been observed historically. USMCA is now subject to an annual joint review by the three governments, meaning that rules of origin can change from year to year. Given that continental supply chains were built under the certainty that these rules would stay in place, firms now have to revisit their compliance decision on a recurring basis instead of making it only once. The framework developed in this paper allows one to understand what each of these policy scenarios may mean for the economy.

In order to do so, I develop a global sourcing model in which heterogeneous firms combine domestic labor with domestically-sourced and imported intermediates, choose the set of countries they source from out of a global menu, and decide, product by product, whether or not to comply with NAFTA/USMCA rules of origin. Compliance grants the firm tariff-free access to the U.S. and Canadian markets, but constrains its sourcing toward North America. Opting out, on the other hand, allows the firm to source globally without any constraints at the cost of paying the MFN tariff, which is the non-discriminatory rate that a WTO member applies to imports from any other WTO member. Given that domestic value added automatically counts toward regional-content requirements, the production structure plays a direct role in the compliance margin. Firms evaluate all of their feasible sourcing strategies and only enter production if the best one yields positive profits. Aggregate income and the

domestic price level adjust in order to clear the Mexican goods market, whereas a competitive non-manufacturing sector pins down the wage and absorbs the reallocation of labor.

I calibrate the model at two different vintages using matched Mexican census-customs firm microdata: a pre-trade-war NAFTA equilibrium (2018 vintage) and a post-USMCA equilibrium (2023 vintage). The main finding of this paper is that USMCA protects Mexico from U.S. tariffs, and that it does so more the higher the tariff on non-compliant goods. With MFN tariffs in place, removing USMCA costs Mexico 1.4% of its real income, whereas at the 25-point surcharge of the 2025 order the loss amounts to 8.9%. With the preferential route available, the same surcharge costs only 0.14%. In other words, what makes the agreement valuable is the outside tariff. For the same reason, content rules cost more when non-compliance is expensive: a 20-point tightening costs about three times as much real income at the 2025 tariff level as at observed tariffs, which is more than the cost of the surcharge itself. The transition from NAFTA to USMCA rules, on the other hand, shows to be close to welfare-neutral, as the two-vintage decomposition attributes $-0.03\%$ of real income to it. When looking at the growth in real income between the two vintages, the largest components that can be computed on a common panel are the growth in foreign demand ($+3.5\%$) and the shift in relative sourcing conditions ($+1.4\%$), under which imports from China rise the most. Around the 2023 baseline, however, content rules are not welfare-neutral. Tightening the regional value content (RVC) threshold by 25 points lowers compliance by 14 points and real income by 0.3%, given that most firms keep complying and buy North American content that they would not otherwise buy. Lastly, a 25% tariff on intermediates from China, a country with which Mexico holds no free trade agreement, cuts Chinese sourcing by 29% at a cost of under a tenth of a percent of real income, since the tariff rebate offsets most of the higher input costs. Asia excluding China is the region that gains the most in proportion ($+2.8\%$), while North American, European and domestic sourcing each rise by 1 to 1.3%.

This paper is related to three strands of the literature. The first is the quantitative global sourcing literature (Antràs et al., 2017, Eaton and Kortum, 2002, Head and Mayer, 2019, Head et al., 2024, Morán and Cebreros, 2025, Tintelnot, 2017), where, typically, firms differ in productivity and countries in their sourcing potential. I build on this framework in two ways. The first one is that I recover, for every firm in the matched census-customs microdata, the parameters behind its sourcing and destination decisions, which are a sourcing efficiency for each source country and a market reach for each destination. Given that these parameters are also filled in for the countries and markets that the firm does not use, both margins are able to move in the counterfactuals. The second one is that compliance is decided product by product, meaning that a multi-product firm can comply on some of its products and pay the tariff on the rest. The closest papers on this margin are Morán and Cebreros (2025)

and Head et al. (2024). The former add a NAFTA-versus-WTO choice with a fixed cost of using NAFTA to the model of Antràs et al. (2017), estimate it on Mexican customs data and target trade flows and firm profits. The latter calibrate a model of the car industry to the cost shares of North American parts and trace the Laffer curve of regional content requirements on employment. In the first paper each firm makes a single NAFTA-or-WTO choice and the sample keeps the products with a CTC rule, whereas in the second compliance is decided per car line under an RVC rule. Neither of them covers the full set of product-level rules of the treaty.

The second strand is the literature on rules of origin in preferential agreements (Conconi et al., 2018, Grossman, 1981, Ju and Krishna, 2002, Ornelas and Turner, 2024). These papers study the rules in theory or measure their effect on trade, and in each of them the rule is a single requirement. In this paper the three compliance regimes of the treaty, change in tariff classification, regional value content and the hybrid rules under which the firm can use either one, enter the same model, with their parameters attached at the HS6 level over the whole manufacturing universe of NAFTA and USMCA. This in turn allows me to take the model to firm-level data for all of manufacturing instead of a single sector.

The third strand is the literature on the reorganization of global supply chains in an environment of rising trade conflicts (Alfaro and Chor, 2025, Fajgelbaum et al., 2024, 2020, Freund et al., 2024, Gopinath et al., 2025, Grossman et al., 2023, Utar, 2026, Utar et al., 2025), which has studied how trade and production reorganize across countries, including their diversion toward Mexico. Notably, these papers do not model whether the diverted trade enters the United States under preference or not, which is the compliance friction at the center of this paper. To this literature I add a structural quantification of Mexican real income under the trade war and the NAFTA-to-USMCA transition taken together, which no structural analysis of USMCA rules of origin with an endogenous compliance margin has reported so far, to the best of my knowledge.

The remainder of this paper is organized as follows. Section 2 documents the empirical patterns in trade flows and USMCA compliance that motivate the model. Section 3 presents the structural framework. I follow by describing the calibration in Section 4 and the estimation results in Section 5. Section 6 presents the counterfactual analysis and Section 7 concludes.

# 2 Motivating Facts

Since 2018, North American manufacturing trade has reorganized substantially in favor of Mexico, while the share of MX→US trade claiming NAFTA/USMCA preference has moved

through three distinct regimes. There are three policy events that plausibly drive this sequence. The first took place between 2018 and 2019, when, in response to the statements of USTR (2018) on forced technology transfer and other unfair trade practices by China, the United States invoked Section 301 of the Trade Act of 1974 in order to impose tariffs on roughly half of its manufactured imports from China. The second was the replacement of NAFTA by USMCA in July 2020. USMCA raised the required North American content for about two fifths of MX→US trade, most of it in the automotive core, and left the rest unchanged or looser (Table 2). The third was the IEEPA executive order of March 2025, a 25% tariff on Mexican imports from which goods that meet USMCA rules of origin are exempt. Figure 1 shows the three trade flows and the compliance rate over this period. Section 2.1 covers the trade flows and Section 2.2 the use of USMCA preference.

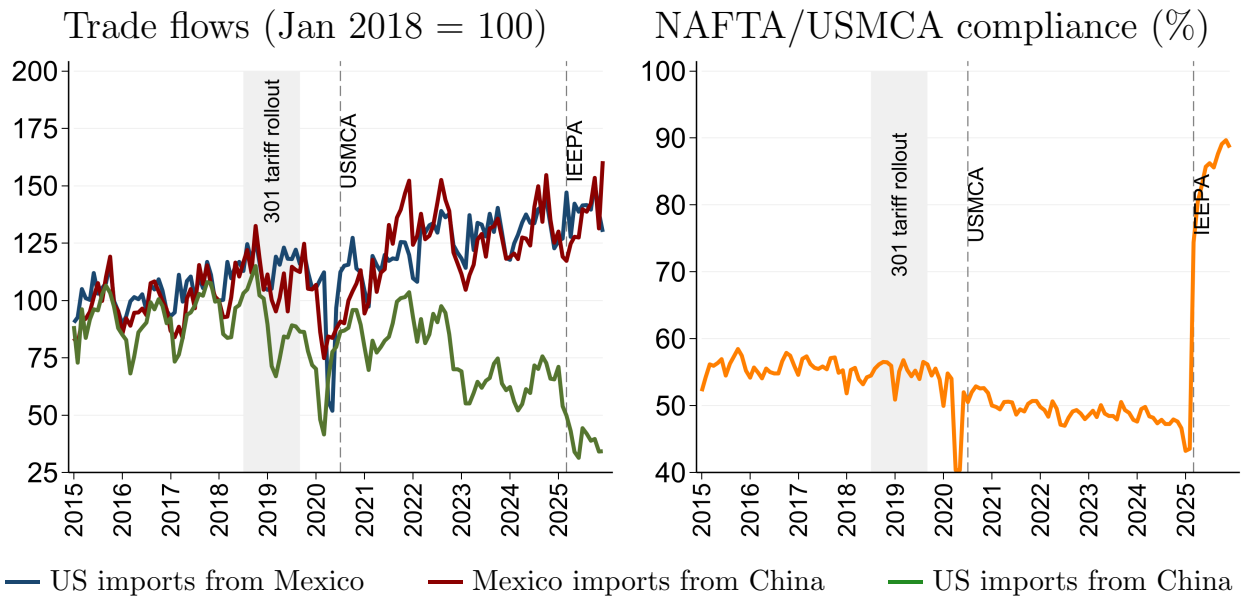



Figure 1: **Trade flows and NAFTA/USMCA compliance, 2015–2025.** Manufacturing HS6 (16–97), deflated to 2023 USD. Monthly through Dec 2025.

## 2.1 The Reorganization of Trade Flows

The first place where Mexico's response to the 2018 trade war can be observed is in trade volumes. U.S. imports from China fell sharply between 2017 and 2025 (−54% in real terms). Mexico took over part of that demand (MX→US: +31%), and at the same time it bought more from China itself (MX←CN: +32%). Most of these additional Chinese purchases are inputs and not consumer goods, as roughly 60% of the MX←CN growth is in intermediate and capital goods[2] and the remaining 40% in final-consumption goods.

[2]I classify goods by end use with the UN Broad Economic Categories, Rev. 5.

A natural question that arises is whether the new MX→US trade took place in the product codes that Mexico was already exporting in 2017 or in newly-emerging product lines, that is, whether the trade response occurred on the intensive or on the extensive margin. The answer shows to be overwhelmingly intensive. Table 1 splits MX→US manufacturing trade by 301 exposure and by whether Mexico already exported the product in 2017. The cell of exposed products that Mexico already exported holds 90% of 2017 trade. The two rows of new products are small. Moreover, 98% of their 2025 value falls in subheadings that only exist since the 2022 revision of the Harmonized System, which by construction have no 2017 trade and no Section 301 coding, so most of that value reflects goods that were already traded under older codes.

In the last column of Table 1 I report the trade-weighted US MFN tariff of each cell, which is the rate a Mexican exporter pays whenever it ships without claiming USMCA preference. These rates are low, at 3.3% on average for the exposed products that Mexico already exported and 3.6% across the four cells. Not claiming preference was therefore not very costly for most shipments, and as I show in Section 2.2, a large part of the eligible trade did not claim it up until the IEEPA order of March 2025, when skipping preference became much more expensive.

Table 1: MX→US manufacturing trade by 301 exposure and pre-shock exporter status, 2017–2025

| | | Trade share (%) | | Trade value ($bn) | | Avg MFN |
|---|---|---|---|---|---|---|
| 301 | Exporter status | 2017 | 2025 | 2017 | 2025 | (%) |
| Exposed | Existing (2017 > 0) | 90.3 | 85.1 | 325.6 | 403.7 | 3.3 |
| Exposed | New (2017 = 0) | 0.0 | 0.1 | 0.0 | 0.4 | 1.2 |
| Not exposed | Existing (2017 > 0) | 9.7 | 9.3 | 35.0 | 44.3 | 6.5 |
| Not exposed | New (2017 = 0) | 0.0 | 5.4 | 0.0 | 25.8 | 7.5 |
| Total | | 100.0 | 100.0 | 360.6 | 474.2 | 3.6 |

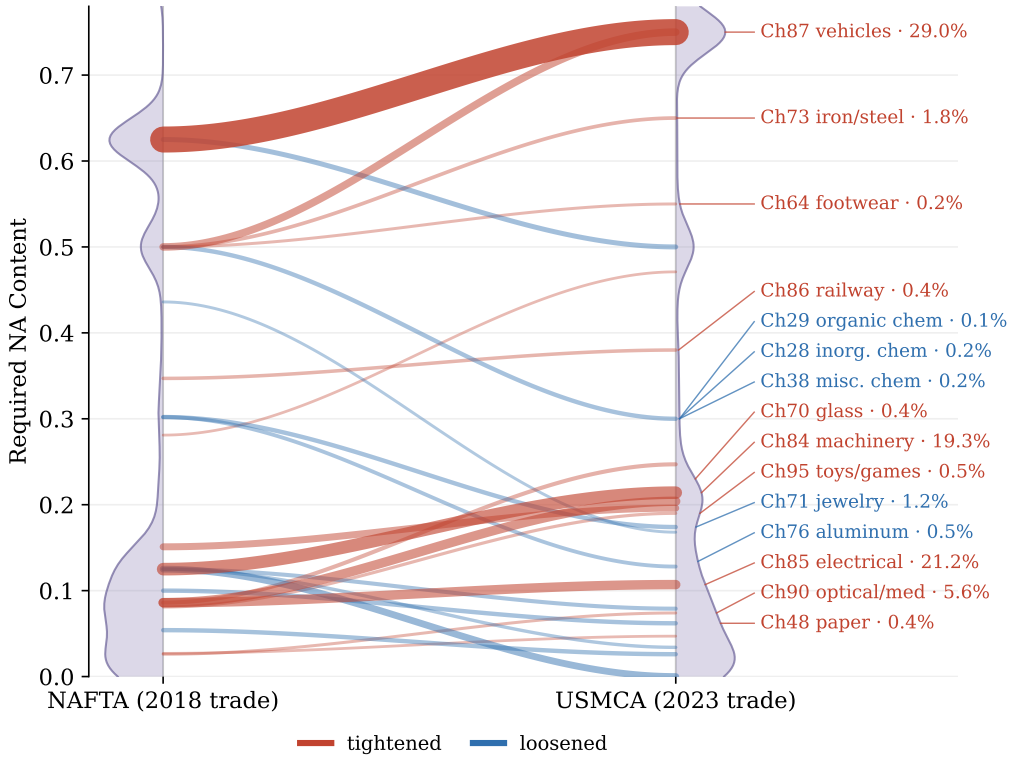
*Notes:* MX→US manufacturing trade (HS 16–97) in 2023 USD, from annual (2017) and monthly (2025) Comtrade data. Exposure to the 301 tariffs follows Bown (2021). MFN rates come from the USITC HTS and are trade-weighted within each cell, although in the (not exposed, new) cell only 0.6% of 2025 value carries a rate.

## 2.2 USMCA Preference Utilization

USMCA inherits the four-regime structure that NAFTA used for its Product Specific Rules of Origin, and adjusts each of these regimes asymmetrically. Under *Change in Tariff Classification (CTC)*, a good qualifies if its non-North American inputs undergo a specified tariff shift through production. *Regional Value Content (RVC)* requires that a minimum share of the good's value originate in North America. *Hybrid* regimes allow a good to qualify

under either CTC or RVC, whichever is easier to satisfy. Lastly, *Yarn-Forward* applies to textiles and requires that upstream yarn originate in North America. This last regime is omitted throughout the paper, given that its multi-stage production requirements go beyond the scope of the quantitative model. The largest change is in the RVC regime. The light-vehicle threshold rises from 62.5% to 75%, phased in over 2020–2023, and two requirements that did not exist under NAFTA are added to it, a Labor Value Content rule (40–45% of value produced by workers earning at least $16/hour) and a 70% North American steel and aluminum purchasing requirement. CTC tightens through HS-specific updates to the tariff shifts, which now tend to require a shift from a more granular subheading. Hybrid rules, which combine the two, inherit both changes.

Figure 2: The USMCA tightening of required North American content is asymmetric




*Notes:* Ribbons bundle HS6 subheadings moving identically NAFTA→USMCA; width proportional to average 2018–2023 MX→US export value. Violins are trade-weighted marginals (2018 for NAFTA, 2023 for USMCA); labels mark the fifteen HS2 chapters with the most trade in subheadings whose requirement moved, with each chapter's average 2018–2023 share of MX→US manufacturing trade.

In Figure 2 I place every manufacturing subheading on a single axis that measures the North American content it requires, and draw each ribbon with a width proportional to average 2018–2023 MX→US exports. For an RVC subheading the position is the threshold written in the rule. For a CTC subheading I compute the floor on the NA content of the final good that the required tariff shift implies, given the subheading's I-O input mix. Hybrid

subheadings are harder to place. I put them on the axis of the parameter that exists under both agreements (the RVC threshold when both carry one, the CTC floor otherwise), and a subheading that is Hybrid under both agreements is drawn on both. The violin on the left shows the trade-weighted distribution of required content under NAFTA and the one on the right under USMCA, while each ribbon follows a group of subheadings that moves together from one agreement to the other.

Most of the mass of the USMCA distribution stays where the NAFTA mass was, and the tightening is concentrated in a set of products that form a second mode near 0.75, which NAFTA did not have. Among the subheadings whose binding parameter is the RVC threshold (RVC and Hybrid), the share of MX→US trade with a required content of 70% or more goes from about 0% under NAFTA to about 30% under USMCA. Almost all of this upper mode is motor vehicles and parts (HS 87), 29% of MX→US manufacturing trade, which converge on the 75% threshold. Machinery (HS 84, 19% of trade) and electrical equipment (HS 85, 21%) account for most of the rest of the increase. About 12% of trade loosens (the blue ribbons).

Table 2 gives the direction of the change within each USMCA regime. In the RVC regime, 64.2% of 2023 trade is in an HS6 whose required content rose and 2.1% in one where it fell, and another 30% came over from CTC at the transition. In CTC about a quarter of the trade tightens, a quarter loosens and nearly half does not move. Hybrid is in between, with 37% tightening and 6% loosening. Weighting the three regimes by 2023 trade, about 40% of manufacturing trade faces a stricter requirement and 12% a looser one, 36% is unchanged and 10% switches between CTC and RVC.

Table 2: Direction of the NAFTA→USMCA rule change, by USMCA regime

| USMCA regime | Within-regime direction of change (% of regime trade) | | | |
|---|---|---|---|---|
| | Tighter ($\Delta > 0$) | Unchanged ($\Delta = 0$) | Looser ($\Delta < 0$) | Regime switch (CTC↔RVC) |
| RVC | 64.2 | 3.3 | 2.1 | 30.4 |
| CTC | 25.5 | 46.7 | 25.9 | 1.9 |
| Hybrid | 36.6 | 57.8 | 5.6 | 0.0 |
| All three (aggregate) | 40.5 | 36.1 | 11.8 | 10.0 |

*Notes:* $\Delta$ from Figure 2. The aggregate row is weighted by 2023 MX→US exports, and the 1.5% of trade that it does not cover is Yarn-Forward. Regime switches are crossings between pure CTC and pure RVC only (products moving into or out of Hybrid are in the tighter, unchanged and looser columns).

Table 3 reports how compliance moved in each of the three main regimes. From 2017 to 2024 the aggregate rate fell by about 8 percentage points, although not by the same amount everywhere, as how much a regime lost depends on whether its binding constraint is a value or a classification. The two regimes whose binding constraint depends on the *value* of inputs absorb almost all of the decline, as RVC falls 14 percentage points (from 94.5% to 80.8%)

Table 3: USMCA compliance regimes: scope, trade share, and compliance change

| | Trade share (%) | | | | Compliance rate (%) | | | | Avg MFN |
|---|---|---|---|---|---|---|---|---|---|
| Regime | 2017 | 2024 | 2025 Jan–Feb | 2025 Mar–Dec | 2017 | 2024 | 2025 Jan–Feb | 2025 Mar–Dec | (%) |
| RVC | 31.6 | 31.4 | 28.3 | 27.5 | 94.5 | 80.8 | 78.2 | 84.4 | 7.6 |
| CTC | 37.3 | 35.9 | 36.3 | 33.0 | 30.8 | 31.6 | 30.3 | 84.3 | 2.6 |
| Hybrid | 29.5 | 31.9 | 34.5 | 38.7 | 46.4 | 34.2 | 29.1 | 85.9 | 1.4 |
| Aggregate | 98.4 | 99.1 | 99.1 | 99.2 | 55.9 | 48.0 | 43.6 | 85.0 | 3.7 |

*Notes:* The Feb/Mar cut isolates the IEEPA order (a 25% tariff on top of the MFN rate on Mexican imports outside USMCA preference, in force from March 4, 2025, with USMCA-compliant goods exempt from March 7). Compliance rates are value-weighted within each regime. Trade shares are computed over the full four-regime manufacturing universe, so what they leave out is the omitted Yarn-Forward regime.

and Hybrid falls 12 percentage points (from 46.4% to 34.2%), whereas CTC barely moves. Given that the trade-weighted MFN on the three modeled regimes was 3.7% (Table 3),[3] the option value of skipping USMCA on eligible trade was small.

This changed in March 2025, when the IEEPA executive order imposed a 25% tariff on Mexican imports on top of the existing MFN rate while exempting goods meeting USMCA rules of origin, which raised the cost of failing to claim preference overnight. The compliance response was both immediate and large. Aggregate value-weighted compliance stayed below its 2024 level in Jan–Feb 2025 (∼44%) and then jumped to ∼85% in Mar–Dec 2025 (Table 3), with CTC and Hybrid both nearly tripling and RVC rising by about six points. I read a jump of this size as evidence that most of the gap before 2025 came from firms that would rather source without restrictions and pay the MFN tariff, and that started claiming preference once that tariff became large enough.

These facts by themselves cannot tell us how costly compliance is for the firms that comply, how much of the decline before 2025 is due to USMCA and how much to the trade war, or what each of these policies would have done on its own. Answering these questions requires a model of how firms decide where to source from and whether to comply.

# 3 Model

There are three things that these facts ask of the model. The first is that claiming preference has to be a decision of the firm, product by product, given that compliance moves within each regime and that not every shipment of the same product carries a claim. The second is that the RVC rule (a minimum North American share of the good's value) and the CTC

[3] Table 1 reports 3.6% because it averages over all MX→US manufacturing trade, including Yarn-Forward and products without a coded rule, weighted by 2017 and 2025 trade, whereas Table 3 covers only the three modeled regimes, weighted by 2017, 2024 and March–December 2025 trade.

rule (non-North American inputs must change tariff class in production) have to enter as separate technologies, since compliance moved differently across regimes in the same way that their rules differ. Lastly, the trade reorganization concentrates in 301-exposed products while Mexico bought more and more Chinese inputs, so firms need to choose their source countries, and where their inputs come from has to matter for whether they can comply. The model can then attribute the observed compliance changes to each policy, and say what compliance would have been under the trade war alone, under USMCA alone, under the higher tariff on non-compliant goods alone, and under all of them.

## 3.1 Environment

I develop a global sourcing model in the tradition of Eaton and Kortum (2002), Antràs et al. (2017), and Tintelnot (2017), which I extend with free trade agreement rules of origin and an endogenous compliance decision at the firm-product level. Firms in the model trade off cheaper inputs from global sourcing against the restrictions they must accept on their supply chain to sell tariff-free in the U.S. and Canada. They take the trade-policy parameters as given.

There is a set of source countries and a set of destination markets. Intermediates are produced in the country blocks

$$s \in \mathcal{S} = \{s_1, s_2, \ldots, s_n\},$$

of which the North American blocks (NA) count as originating under the agreement. Output is sold in the destination blocks

$$d \in \mathcal{D} = \{d_1, d_2, \ldots, d_m\},$$

which include the domestic market, Mexico (MX), as well as the U.S. and Canada (U.S. & CA), which is the only destination where the rules of origin apply. Note that the mechanisms of the model do not depend on the exact composition of $\mathcal{S}$ and $\mathcal{D}$. The blocks that are used in the estimation are listed in Section 4.

## 3.2 Consumers

Consumers in each destination $d$ split total expenditure $Y_d$ between manufactures and a homogeneous non-manufacturing good $C_{o,d}$ through a Cobb-Douglas upper tier,

$$U_d = C_{M,d}^{\chi_d} C_{o,d}^{1-\chi_d}, \qquad \chi_d \in (0,1), \tag{1}$$

where $C_{M,d}$ is the manufacturing composite and the non-manufacturing good is the numeraire ($p_o \equiv 1$). A constant share of expenditure falls on manufactures,

$$E_d \;\equiv\; \chi_d\, Y_d, \tag{2}$$

with the remainder $(1-\chi_d)Y_d$ spent on the non-manufacturing good. The manufacturing composite is itself a CES aggregate over a continuum of differentiated varieties $\omega \in \Omega_d$, with exogenous appeal weights $b_d(\omega) > 0$ scaling effective consumption:

$$C_{M,d} = \left( \int_{\omega \in \Omega_d} \left( b_d(\omega)\, q_d(\omega) \right)^{\frac{\sigma-1}{\sigma}} d\omega \right)^{\frac{\sigma}{\sigma-1}}, \tag{3}$$

with $\sigma > 1$ the elasticity of substitution. Utility maximization subject to the manufacturing budget $\int p_d(\omega)\, q_d(\omega)\, d\omega = E_d$ yields the quantity demanded

$$q_d(\omega) \;=\; E_d\, P_d^{\sigma-1}\, p_d(\omega)^{-\sigma}\, b_d(\omega)^{\sigma-1}, \tag{4}$$

where $P_d = \left( \int \left( p_d(\omega)/b_d(\omega) \right)^{1-\sigma} d\omega \right)^{1/(1-\sigma)}$ is the CES price index of destination $d$. The only difference across destinations lies in what is endogenous. For the foreign markets $d \in \{\text{U.S. \& CA, RoW}\}$, total expenditure $Y_d$ is exogenous, so $E_d$ is taken as given and the share $\chi_d$ is absorbed into its calibrated level. For Mexico, on the other hand, manufacturing non-wage income $\Pi_M$ is endogenous and non-manufacturing income $\Pi_o$ is exogenous (both are defined in Section 3.4), so $E_M$ moves with Mexican income $w_m \bar{L} + \Pi_M + \Pi_o$ through (34), which is what transmits the feedback from firm activity to domestic demand through $\Pi_M$.

## 3.3 Firms

**Primitives.** Each potential firm $f$ in mass $\bar{M}$ is characterized by three objects: a product set $\mathcal{P}_f$ (empirically set at the HS6 level), a vector of country-specific sourcing efficiencies $\{a_{f,s}\}_{s \in \mathcal{S}}$, and a vector of destination-specific market reach $\{b_{f,d}\}_{d \in \mathcal{D}}$. The sourcing efficiency $a_{f,s}$ captures the firm's supplier networks and institutional relationships in source country $s$. For instance, U.S.-owned firms tend to have a high $a_{f,\text{U.S. \& CA}}$, Chinese-owned firms tend to have a high $a_{f,CH}$, and so on. The market reach $b_{f,d}$ plays the analogous role on the demand side, scaling revenue at destination $d$ in order to capture features such as market access, distribution networks, and brand recognition. Each product $p \in \mathcal{P}_f$ in turn determines the compliance regime and the MFN tariff that the firm faces.

**Sourcing Strategy.** Each firm sources a continuum of intermediate inputs $j \in [0,1]$ from source countries in $\mathcal{S}$. To source from country $s$, the firm knows how well it can source

from it through $a_{f,s}$ and pays a fixed cost $w_m \kappa_s \geq 0$ in units of Mexican labor. Let $\mathcal{S}_f \subseteq \mathcal{S}$ denote the set of countries from which the firm sources. I refer to $\mathcal{S}_f$ as the firm's *sourcing strategy*, and assume that it is shared across all of the firm's products and destinations.[4]

Conditional on paying the fixed cost, the firm draws input-level productivities $z_s(j) \sim$ Fréchet$(T_s a_{f,s}^{\theta}, \theta)$ independently across inputs and source countries. These draws capture every firm-, input- and country-specific factor that plays a role in the realized landed cost, such as the quality of the supplier match, the reliability of shipping or how well the product fits. $T_s > 0$ is the absolute sourcing advantage of country $s$, $\theta$ governs the dispersion of the productivity draws, and the firm-country sourcing efficiency $a_{f,s}$ shifts the location of the distribution upward, which raises the likelihood that country $s$ offers a lower landed cost for a given input. The per-input landed cost is

$$c_{f,s}(j) = \frac{c_s \tau_s}{z_s(j)} \tag{5}$$

where $c_s$ is the cost of inputs in source country $s$ and $\tau_s$ the iceberg trade cost of sourcing from $s$. The firm then picks, for each input $j$, the source that offers it at the lowest cost, $s^*(j) = \arg\min_{s \in \mathcal{S}_f} c_{f,s}(j)$. Integrating these per-input minima over the continuum of inputs gives the intermediate cost index, which under the Fréchet distribution has the following closed form:

$$\begin{aligned} c_f^M(\mathcal{S}_f) &= \int_0^1 \min_{s \in \mathcal{S}_f} \frac{c_s \tau_s}{z_s(j)} \, dj \\ &= \Gamma\left(1 - \frac{1}{\theta}\right) \left( \sum_{s \in \mathcal{S}_f} T_s a_{f,s}^{\theta} (c_s \tau_s)^{-\theta} \right)^{-1/\theta}. \end{aligned} \tag{6}$$

The expected share of intermediates sourced from country $s$ is the integral over $j$ of the indicator that $s$ offers the minimum landed cost, which the same algebra evaluates as

$$\begin{aligned} \mu_{f,s}(\mathcal{S}_f) &= \int_0^1 \mathbf{1}\left\{ \frac{c_s \tau_s}{z_s(j)} = \min_{s' \in \mathcal{S}_f} \frac{c_{s'} \tau_{s'}}{z_{s'}(j)} \right\} dj \\ &= \frac{T_s a_{f,s}^{\theta} (c_s \tau_s)^{-\theta}}{\sum_{s' \in \mathcal{S}_f} T_{s'} a_{f,s'}^{\theta} (c_{s'} \tau_{s'})^{-\theta}}. \end{aligned} \tag{7}$$

**Production Technology.** Each active firm produces output by combining Mexican

[4] Customs data record input origin and output destination separately, but not which input made which output, so a shared $\mathcal{S}_f$ is what the data support.

labor $\ell_f$ with an intermediate composite $M_f$ through a Cobb-Douglas technology:

$$q_f = \ell_f^{\gamma} M_f^{1-\gamma}, \tag{8}$$

where $\gamma \in (0,1)$ is the value-added share of labor. Labor is hired locally at wage $w_m$, and its cost in turn counts as North American value added by construction. The intermediate composite $M_f$ aggregates a continuum of inputs $j \in [0,1]$, each sourced from the lowest-cost country in the firm's sourcing set. Note how this allows the possibility for a firm to comply with the trade agreement requirements not only by sourcing inputs from North America, but also through the value added by Mexican labor.

Combining (8) with (6), cost-minimization over labor and the intermediate composite yields the Cobb-Douglas unit cost:

$$c_f(\mathcal{S}_f) = B\, w_m^{\gamma} \left[c_f^M(\mathcal{S}_f)\right]^{1-\gamma}, \tag{9}$$

where $B \equiv \gamma^{-\gamma}(1-\gamma)^{-(1-\gamma)}$ is a constant derived from the cost-minimization problem.

**Compliance: Three Regimes.**[5] Rules of origin bind only on U.S. & CA-bound output. For each product $p \in \mathcal{P}_f$, the firm chooses a compliance indicator $d_{fp} \in \{0,1\}$, where $d_{fp} = 1$ means that the firm accepts the applicable input restriction in exchange for tariff-free access to the U.S. & CA market, whereas $d_{fp} = 0$ means that it pays the MFN tariff $\tau_p^{MFN}$ on U.S. & CA-bound shipments and sources freely. The different regimes are quantified as follows:

- **Change in Tariff Classification (CTC).** A CTC rule for output $p$ names, at a treaty-determined HS depth (chapter, heading or subheading), a specific set $\mathcal{P}(p)$ of input HS classes whose tariff classification must change *through production* in order for the input to qualify. A non-NA input from $\mathcal{P}(p)$ whose tariff class is unchanged through production fails the rule, whereas an NA-sourced input always passes, regardless of its classification. I take the I-O technology at output $p$ as binding, meaning that any input in $\mathcal{P}(p)$ to which the I-O table assigns a positive expenditure share at $p$ is interpreted as failing the classification-change test under $p$'s production technology, and must therefore be NA-sourced for compliance. The rule then reduces to a single product-specific scalar, where $\phi_p(j)$ is the I-O expenditure share of input class $j$ in $p$'s intermediate bundle:

$$\alpha_p \;=\; \sum_{j \in \mathcal{P}(p)} \phi_p(j),$$

[5] Yarn-Forward is omitted given that textiles and apparel account for under 2% of MX→US manufacturing trade, and that their multi-tier yarn→fabric→garment origin chain goes beyond the model's single-tier sourcing.

which is the I-O share of $p$'s intermediate bundle allocated to those restricted-and-unchanged classes. Through the lens of the model, the rule maps to a subset of the firm's continuum of input varieties $j \in [0,1]$ of measure $\alpha_p$ that must be NA-sourced, while the residual measure $1-\alpha_p$ is drawn from the unrestricted set $\mathcal{S}_f$. Given that input varieties are drawn iid Fréchet across countries, which varieties end up in the restricted subset does not matter for its cost, only its measure does, meaning that the compliant intermediate cost takes the form of a Cobb-Douglas aggregation between the restricted (NA-only) and the unrestricted sub-continua:

$$c_{f,p}^{M,CTC}(\mathcal{S}_f) = \Gamma\big(1-\tfrac{1}{\theta}\big)\left(\sum_{s\in NA\cap\mathcal{S}_f} T_s a_{f,s}^{\theta}(c_s\tau_s)^{-\theta}\right)^{-\alpha_p/\theta}\left(\sum_{s\in\mathcal{S}_f} T_s a_{f,s}^{\theta}(c_s\tau_s)^{-\theta}\right)^{-(1-\alpha_p)/\theta}. \tag{10}$$

Both trade agreements further allow a *de minimis* relaxation, under which CTC is treated as satisfied whenever non-conforming non-originating inputs account for at most $\delta_p = 10\%$ of the good's value under USMCA and 7% under NAFTA. Given that $\alpha_p$ is a share of intermediate spending and that intermediates make up a fraction $1-\gamma$ of the good's cost, the tolerance trims the binding restricted measure to $\mu_p^{CTC} = \max\big(\alpha_p - \delta_p/(1-\gamma),\, 0\big)$, where the good's value is measured at total cost, which is the same basis as the net-cost RVC thresholds below. This effectively allows firms to have a small margin of deviation from the CTC requirement without losing preference eligibility.

- **Regional Value Content (RVC).** The North American content share of production must reach a product-specific threshold $\mu_p^{RVC}$. Given that Mexican labor counts as North American content, the content share under an unrestricted sourcing strategy is

$$\gamma + (1-\gamma)\sum_{s\in NA\cap\mathcal{S}_f} \mu_{f,s}(\mathcal{S}_f) \geq \mu_p^{RVC}, \tag{11}$$

where the first term is the automatic contribution of Mexican labor and the second is the NA share of the firm's intermediate sourcing. The rule is met when the value added both from labor and North American manterial exceeds the threshold determined by the free trade agreement.

- **Hybrid.** The firm may qualify under either rule and takes the cheaper compliant cost:

$$c_{f,p}^{Hyb}(\mathcal{S}_f) \;=\; \min\big(c_{f,p}^{CTC}(\mathcal{S}_f),\; c_{f,p}^{RVC}(\mathcal{S}_f)\big)\,.$$

Note that the firm decides on compliance separately for each product and destination (only U.S. & CA-bound shipments face the rules), while its sourcing strategy $\mathcal{S}_f$ is the same for all of its products and destinations. A firm can therefore comply for one product and not for another, choosing its compliance vector $\{d_{fp}\}$ product by product, but once it has paid the sourcing fixed costs it buys from the same countries for every product. I keep a single sourcing strategy per firm for two reasons. It keeps the discrete-choice problem tractable, and it is what the customs data allow, as they record the import composition of the firm but not which inputs go into which product.

Conditional on $\mathcal{S}_f$ and the per-product compliance indicator $d_{fp}$, the destination-specific landed unit cost is

$$c_{f,p,d}(\mathcal{S}_f, d_{fp}) \;=\; \begin{cases} c_{f,p}(\mathcal{S}_f) & \text{if } d = \text{U.S. \& CA and } d_{fp} = 1, \\ c_f(\mathcal{S}_f)\,(1+\tau_p^{MFN})^{\sigma/(\sigma-1)} & \text{if } d = \text{U.S. \& CA and } d_{fp} = 0, \\ c_f(\mathcal{S}_f) & \text{if } d \in \{\text{MX, RoW}\}, \end{cases} \tag{12}$$

where $c_f(\mathcal{S}_f)$ is the unrestricted unit cost from (9) and $c_{f,p}(\mathcal{S}_f)$ the compliant unit cost, which is $c_{f,p}^{CTC}(\mathcal{S}_f)$, $c_{f,p}^{RVC}(\mathcal{S}_f)$ or $c_{f,p}^{Hyb}(\mathcal{S}_f)$ depending on the regime that governs product $p$. Note that the choice $d_{fp} = 1$ requires at least one North American country in $\mathcal{S}_f$ under every regime, given that both the CTC restriction $\alpha_p$ and the RVC restriction $\alpha_{f,p}^*$ draw the restricted inputs from NA sources. Specifically, a firm that complies produces at the restricted cost $c_{f,p}$ and does not pay a tariff, whereas a firm that does not comply produces at the unrestricted cost $c_f$ but pays the MFN tariff on its U.S. & CA-bound revenue. The reason behind the exponent $\sigma/(\sigma-1)$ is that the duty is ad valorem on the fob value, and this exponent is what converts it into a wedge on cost. The U.S. & CA buyer pays $p(1+\tau_p^{MFN})$ and the firm receives $p = \frac{\sigma}{\sigma-1}c_f$, while the duty $\tau_p^{MFN}\, p\, q$ goes to the U.S. Treasury. The fob revenue and profit of a non-compliant product then scale as $(1+\tau_p^{MFN})^{-\sigma}$, which is exactly what the wedge gives in (13). MX and RoW shipments are unaffected by the compliance choice.

**Destination Markets.** Each firm chooses a destination set $\mathcal{D}_f \subseteq \mathcal{D}$, and pays a fixed entry cost $w_m \kappa_d$ in units of Mexican labor for each $d \in \mathcal{D}_f$. Each product of an active firm is a differentiated variety at every destination that it serves. Given constant-markup pricing over the duty-free unit cost, the firm's fob revenue from product $p$ at destination $d$ is

$$R_{f,p,d}(\mathcal{S}_f, b_{f,d}) \;=\; \varpi_{f,p}\, E_d\, P_d^{\sigma-1} \left(p_{f,p,d}(\mathcal{S}_f, d_{fp})\right)^{1-\sigma} b_{f,d}^{\sigma-1}, \tag{13}$$

where $p_{f,p,d}(\mathcal{S}_f, d_{fp}) = \frac{\sigma}{\sigma-1}\, c_{f,p,d}(\mathcal{S}_f, d_{fp})$ is the constant-markup price over the destination-

specific landed unit cost from (12), and $\varpi_{f,p}$ is product $p$'s share of the firm's observed exports times its number of products, so that the weights average to one within the firm.

**Profits and the Firm Problem.** The firm jointly chooses its sourcing set $\mathcal{S}_f$, its destination set $\mathcal{D}_f$ and a per-product compliance vector $\{d_{fp}\}_{p\in\mathcal{P}_f}$ in order to maximize expected profits:

$$\begin{aligned}\pi_f = \max_{\mathcal{S}_f,\,\mathcal{D}_f,\,\{d_{fp}\}} \; & \frac{1}{\sigma}\sum_{d\in\mathcal{D}_f}\sum_{p\in\mathcal{P}_f} R_{f,p,d} \;-\; w_m\sum_{s\in\mathcal{S}_f}\kappa_s \;-\; w_m\sum_{d\in\mathcal{D}_f}\kappa_d \\ \text{subject to } \; & \mathcal{S}_f\subseteq\mathcal{S}, \quad \mathcal{D}_f\subseteq\mathcal{D}, \quad d_{fp}\in\{0,1\}, \\ & d_{fp}=1 \implies \begin{cases} \mu_p^{RVC}\geq\mu_p^{CTC} & p\in\text{CTC}, \\ \gamma+(1-\gamma)\mu_f^{NA}(\mathcal{S}_f)\geq\mu_p^{RVC} & p\in\text{RVC}, \\ \text{either condition} & p\in\text{Hybrid}\end{cases}\end{aligned} \tag{14}$$

where $\mu_f^{NA}(\mathcal{S}_f)\equiv\sum_{s\in NA\cap\mathcal{S}_f}\mu_{f,s}(\mathcal{S}_f)$ is the North American share of the firm's intermediate bundle. Since every product and destination of the firm draws on the same sourcing set, the firm has only one such share. It chooses its North American content for its input pool as a whole, and not product by product. A Hybrid product qualifies under either rule.

As in Antràs et al. (2017), solving (14) is a combinatorial problem, in which the firm evaluates every non-empty sourcing set against every destination set and selects the pair that yields the highest profit, with the additional layer that, for each pair, it decides product by product whether to comply (Appendix A.5).

**Endogenous Entry and Exit.** Let $\pi_f(w_m)$ denote the maximized expected profit from (14), with the dependence on the Mexican wage made explicit. Firm $f$ is *active* if and only if some non-empty $(\mathcal{S}_f,\mathcal{D}_f)$ yields positive profits:

$$f\in\mathcal{A}(w_m) \quad\Longleftrightarrow\quad \pi_f(w_m)>0. \tag{15}$$

The endogenous active set $\mathcal{A}(w_m)\subseteq[0,\bar{M}]$ is the subset of potential firms for whom some sourcing, destination and compliance strategy is profitable. Inactive firms do not produce, source inputs, or employ Mexican labor. As in Tintelnot (2017), the mass of potential firms $\bar{M}$ is taken as a primitive, so that there is no free-entry condition. Considering that firm heterogeneity is multidimensional (the full sourcing-efficiency vector $\{a_{f,s}\}$ and market-reach vector $\{b_{f,d}\}$), there is no single productivity cutoff separating active from inactive firms.

## 3.4 Market Clearing and Equilibrium

Mexico is endowed with a fixed labor force $L$, which is mobile across the manufacturing sector modeled above and a homogeneous non-manufacturing sector. The non-manufacturing sector is perfectly competitive and produces the numeraire good with labor alone under constant returns, $Q_o = A_o L_o$, where $A_o$ is its labor productivity. With free entry, price equals marginal cost, $p_o = w_m/A_o$, and since the non-manufacturing good is the numeraire ($p_o \equiv 1$), the Mexican wage is tied to productivity in that sector,

$$w_m = A_o. \tag{16}$$

Manufacturing thus hires at a wage that it cannot move, since workers can always earn $w_m$ in an outside sector that is competitive and large enough to take them in, and labor moves between the two sectors without changing it. Antràs et al. (2017) close their model in the same way.[6]

The labor force $L$ then divides itself between the two sectors. Manufacturing firms hire as many workers as they want at $w_m = A_o$, and the rest, $L_o = L - L_m \geq 0$, work in the non-manufacturing sector. Letting $VC_f \equiv \frac{\sigma-1}{\sigma}\sum_{d,p} R_{f,p,d}$ denote total variable cost, manufacturing labor demand aggregates over the active set $\mathcal{A} \equiv \mathcal{A}(w_m)$:

$$L_m = \int_{f\in\mathcal{A}} \Big[ \big(\gamma + (1-\gamma)\mu_{f,MX}\big)\tfrac{VC_f}{w_m} \;+\; \sum_{s\in\mathcal{S}_f} \kappa_s \;+\; \sum_{d\in\mathcal{D}_f} \kappa_d \Big] df, \tag{17}$$

where the bracket adds up production labor (the $\gamma$ share of variable cost under Cobb-Douglas), Mexican-sourced intermediates (the $(1-\gamma)\mu_{f,MX}$ share) and the fixed costs of sourcing and destination entry. Note that no wage is solved for through equation (17). I use it as an accounting identity that reads $L_m$ off the firm-level solution, and the pinned-wage closure is interior as long as $L_m \leq L$.

For the foreign destinations $d \in \{\text{U.S. \& CA, RoW}\}$, manufacturing expenditure and prices $(E_d, P_d)$ are taken as exogenous, considering that Mexico is a small open economy relative to those markets and its firms' decisions are not expected to materially move external incomes or price levels. The Mexican pair $(E_M, P_M)$, on the other hand, is determined within the model, as described in Appendix A.8.

The Mexican price index $P_M$ in turn aggregates the prices of Mexican-firm varieties sold

[6] Non-manufacturing accounts for about 75% of Mexican employment (INEGI, ENOE), which supports the perfectly-elastic labor-supply pole.

at MX through the CES dual of (3):

$$\left(P_M\right)^{1-\sigma} = \int_{f\in\mathcal{A}:\, MX\in\mathcal{D}_f} \sum_{p\in\mathcal{P}_f} \left(\frac{p_{f,p,MX}}{b_{f,MX}}\right)^{1-\sigma} df. \tag{18}$$

The welfare metric that I report in the counterfactuals is Mexican consumer real income under the Cobb-Douglas upper tier. The ideal price index dual to $U = C_M^{\chi} C_o^{1-\chi}$ is the geometric mean $P \propto P_M^{\chi}\, P_o^{1-\chi}$, so deflating aggregate nominal Mexican income by $P$ gives

$$W_M \;=\; \frac{w_m L \;+\; \Pi_M + \Pi_o}{P_M^{\chi}\, P_o^{1-\chi}}, \tag{19}$$

where the three numerator components are, in order, the economy-wide wage bill (both sectors pay $w_m = A_o$ over the fixed labor supply $L$), manufacturing non-wage income

$$\Pi_M \;=\; \int_{f\in\mathcal{A}} \pi_f \, df, \tag{20}$$

and non-manufacturing income $\Pi_o$. Given that this sector is perfectly competitive and operates under constant returns, free entry drives economic profits to zero, so that $\Pi_o \equiv 0$. Taking $P_o = 1$ as the numeraire, the denominator of (19) reduces to $P_M^{\chi}$. Given that $w_m = A_o$ and $L$ do not move under the pinned-wage closure, any movement in the numerator of the welfare measure has to come through $\Pi_M$.

Putting all of the above together, I can now define the equilibrium of the model.

Given preference $(\sigma, \chi)$, technology $(\theta, \gamma, A_o)$, firm $(\mathcal{P}_f, \{a_{f,s}\}, \{b_{f,d}\})$, policy $(\{\alpha_p, \mu_p^{RVC}, \tau_p^{MFN}\})$, foreign-market $(\{E_d, P_d\}_{d\neq MX},\ \{T_s, c_s, \tau_s\})$, and fixed-cost $(\{\kappa_s, \kappa_d\})$ primitives, together with the Mexican labor force $L$, an equilibrium is a tuple

$$\left(\mathcal{A},\ \{\mathcal{S}_f, \mathcal{D}_f, (d_{fp})_{p\in\mathcal{P}_f}\}_{f\in\mathcal{A}}\right)$$

such that, at the pinned wage $w_m = A_o$:

(i) Consumers satisfy the Cobb-Douglas upper tier (1) and CES demand (4) at each destination.

(ii) Each active firm $f \in \mathcal{A}$ solves (14) for $(\mathcal{S}_f, \mathcal{D}_f, \{d_{fp}\})$, taking the domestic pair $(E_M, P_M)$ and the foreign pairs $\{(E_d, P_d)\}_{d\neq MX}$ as given.

(iii) Entry satisfies (15): $\mathcal{A} = \{f : \pi_f(w_m) > 0\}$.

(iv) Domestic demand is consistent: $E_M$ satisfies (34) with $\Pi_M$ from (20), and $P_M$ satisfies (18), both at the firms' choices.

Given that the wage is pinned, solving for the equilibrium only requires iterating on the domestic pair $(E_M, P_M)$ until condition (iv) holds, which I describe in Appendix A.8.

# 4 Calibration

In the calibration, the sourcing side is divided into six country blocks,

$$\mathcal{S} = \{\text{U.S. \& Canada (U.S. \& CA), Mexico (MX), Europe (EU),} \\ \text{China (CH), Asia excl. China (AX), Rest of World (RoW)}\},$$

and the sales side into three destination blocks,

$$\mathcal{D} = \{\text{U.S. \& Canada (U.S. \& CA), Mexico (MX), Rest of World (RoW)}\}.$$

I pool the U.S. and Canada on both the sourcing and the demand side. The two share a compliance regime, and from Mexico's perspective they play the same role in the model, that of a single tariff-free North American partner. China is kept apart from the rest of Asia in order to be able to tariff a non-FTA partner on its own. Having separate destinations in turn allows the firm to decide on compliance product by product according to the share of its revenue that goes to the U.S. & CA, instead of taking an all-or-nothing decision at the plant level.

## 4.1 Data

I take the model to firm-level Mexican data drawn from two sources: the *Profile of Export Manufacturing Enterprises* (PEME) customs microdata and the Mexican *Economic Census* (CE). PEME records the universe of Mexican customs transactions at the firm $\times$ HS8 $\times$ partner $\times$ year level, and from it I construct firm-level import-source shares $\mu_{f,s}$ for $s \neq$ MX and export-destination shares $\omega_{f,d}$ for $d \in \{\text{U.S. \& CA, RoW}\}$. From the CE, which covers the population of Mexican manufacturing establishments in 2019 and 2024, I recover each firm's domestic material purchases, domestic sales, employment and paid wage bill, ownership and industry. Given that the CE is collected at the establishment level, I add up the establishments of each firm using the INEGI firm identifier, and I take the industry and ownership of the firm from its largest establishment. This same identifier is also available in PEME, which allows me to link the two datasets and to observe, for every matched firm, both its census characteristics and its customs sourcing and destination shares. Since each census asks about the previous year, the 2019 and 2024 censuses are matched to PEME 2018

and 2023, with every peso magnitude of the 2018 vintage expressed in constant 2023 pesos (the matches are described in Appendix C).

The firms that enter the model individually are the census firms that match a PEME firm and that trade on at least one side, whether it is by selling abroad or by importing materials. This means that import-only firms, which buy abroad but do not export, are also part of the model. I give each of these firms one potential export product, namely the modal HS6 of its six-digit SCIAN class in the Pierce and Schott (2012) concordance weighted by Mexican exports to the U.S. and Canada. If in a counterfactual its imputed market reach takes it into the U.S. & CA market, the firm then faces the rule and the MFN rate of that product. All the other census firms, those without a customs record and those with one but no trade, are collapsed into bins by NAICS-3 industry. They come in all sizes and account for a non-trivial share of manufacturing activity, so they still matter for the Mexican price index, where they enter as an additive term $K$ in the CES sum, $P_M^{1-\sigma} = T + K$. Here $T$ aggregates the price terms of the traders and $K$ is calibrated to the baseline share of the non-traders. Since I model the non-traders as sourcing and selling only at home at the pinned wage, their prices do not move in the counterfactuals and neither does $K$, but $K$ still dampens the response of $P_M$, given that the traders only make up part of the CES sum. The sales of the non-traders, on the other hand, do move with Mexican demand, and their variable profits (a fraction $1/\sigma$ of those sales) are part of manufacturing non-wage income $\Pi_M$.

The firm-level block runs on the traders, 5,719 firms in 2018 and 5,133 in 2023. Table 4 lists the parameters, which I calibrate in two layers. Some are fixed outside of the model, either from data or from the literature. The rest are structural, and I estimate them through model inversions, Heckman-corrected regressions and a moment-inversion estimator that matches aggregate participation shares.

## 4.2 Externally Calibrated Parameters

**Preferences ($\sigma$).** Following Antràs et al. (2017), I pin the CES elasticity $\sigma$ from the firm's first-order condition under monopolistic competition, under which revenue equals a constant markup over variable cost, $R_f = \frac{\sigma}{\sigma-1} VC_f$, with $VC_f = wL_f + M_{t,f}$. I take the median of the firm-level markup $R_f/VC_f$ across all CE manufacturing firms in each census wave, pool the two medians with weights proportional to each wave's total revenue, and invert the pooled markup, so that the same $\sigma$ enters both vintages. Pooling is a deliberate choice, as it imposes that $\sigma$, a deep preference parameter, is constant over the five-year window, meaning that the decomposition in Section 6 is attributable to the inputs that do vary. The median markup is 2.00 in the 2019 wave and 1.82 in the 2024 wave, and the pooled markup is 1.91, which inverts to $\sigma = 2.10$.

Table 4: Calibration: Parameter Values and Identification

| **Parameter** | **Object** | **Identified from** |
|---|---|---|
| *A priori / data* | | |
| $\sigma$ | CES elasticity | Unweighted median firm markup $R_f/VC_f = \sigma/(\sigma-1)$ (CE) |
| $\gamma_s$ | Labor share | Sector (NAICS-3)cost share $\sum wL/\sum(wL+M_t)$ (CE) |
| $\alpha_p,\ \mu_p^{RVC}$ | Product RoO parameters | NAFTA/USMCA + INEGI I-O |
| $\tau_p^{MFN}$ | MFN tariff | USITC |
| $E_d,\ P_d,\ d \in$ {U.S. & CA, RoW} | Foreign mfg. expenditure / price | World Bank WDI |
| $\psi_s \equiv T_s(c_s\tau_s)^{-\theta}$ | Country sourcing shifter | Mean firm sourcing shares (CE and PEME) |
| $L$ | Mexican labor force | ENOE economically active population |
| $A_o$ | Non-mfg. productivity | Annual wage per worker (CE) |
| $\theta = 1.79$ | Fréchet shape | Antràs et al. (2017) |
| *Structural* | | |
| $a_{f,s}$ (per source country) | Firms' sourcing efficiency | Model based inversion and Heckman two-step over missing cells |
| $b_{f,d}$ (per destination) | Firms' market reach | Model based inversion and Heckman two-step over missing cells |
| $\kappa_s$ (per source) | Source-entry fixed cost | Moment inversion on source participation shares |
| $\kappa_d$ (per destination) | Destination-entry fixed cost | Moment inversion on destination participation shares |
| *Equilibrium* | | |
| $L_m$ | Manufacturing employment | Labor allocation (17) (non-mfg. residual) |
| $E_M,\ \Pi_M,\ P_M$ | Equilibrium MX expenditure / income / price | Domestic-demand fixed point (34), (18) and (20) |

**Technology ($\gamma$).** Under Cobb-Douglas production, the labor share of variable cost equals the production-function exponent (Appendix A.2), which justifies calibrating $\gamma$ as a cost share. Given that input intensity differs sharply across manufacturing branches, I compute the share at the NAICS-3 sector level as a ratio of sums, $\gamma_s = \sum wL / \sum(wL + M_t)$, where the paid wage bill $wL$ and own-materials purchases $M_t$ are aggregated across *all* CE manufacturing firms in sector $s$ (and not only across the traders of the model block), and then assign each model firm its sector's value. The sector shares are computed within each census wave and then frozen at their simple mean across the two waves. Appendix Figure 8 shows the per-wave and pooled values across the 21 NAICS-3 sectors. When weighted by the wage bill of the estimation panels, the pooled shares aggregate to 0.204 (2018) and 0.207 (2023).

**Dispersion ($\theta$).** The Fréchet shape governing the dispersion of firm-level sourcing efficiency is set to $\theta = 1.79$ following Antràs et al. (2017).

**Foreign-market block: country sourcing shifters ($\psi_s$).** The bundle $\psi_s \equiv T_s(c_s\tau_s)^{-\theta}$ is identified from the mean sourcing shares of the firms. Given that $T_s$, $c_s$ and $\tau_s$ cannot be identified separately, I work with $\psi_s$ as a whole and normalize $\psi_{\text{U.S. \& CA}} \equiv 1$. The reason behind taking U.S. & CA as the normalizing coordinate is that it is the non-home block with the highest participation, which in turn provides a stable non-zero denominator for the firm-level ratio $\mu_{f,s}/\mu_{f,\text{U.S. \& CA}}$, which the structural step below relies on. Setting $a_{f,s} \equiv 1$ at this stage, which is without loss of generality, reduces (7) to $\bar{\mu}_s/\bar{\mu}_{\text{U.S. \& CA}} = \psi_s$, and each remaining $\hat{\psi}_s$ is then simply a ratio of sample means of the CE–PEME sourcing shares. I follow by restoring the firm-specific $a_{f,s}$ in Section 4.3.

**Foreign-market block: manufacturing expenditure and prices ($E_d$, $P_d$).** The CES revenue equation $R_{f,p,d} = \varpi_{f,p} E_d P_d^{\sigma-1} p_{f,p,d}^{1-\sigma} b_{f,d}^{\sigma-1}$ takes the manufacturing expenditure $E_d$ and the price $P_d$ as exogenous on the foreign side, with the upper-tier share $\chi_d$ absorbed into $E_d$ (2). I normalize $E_d \equiv P_d \equiv 1$ in the 2018 vintage for both $d \in \{\text{U.S. \& CA, RoW}\}$. For the 2023 vintage, I take the 2018→2023 growth of nominal GDP and of the implicit GDP deflator from the World Bank WDI and set $E_d$ to their ratio. The countries are aggregated following the three destinations of the model, with U.S. & CA = USA + CAN and RoW = WLD − USA − CAN − MEX. Unlike the foreign pairs, the Mexican pair is not anchored to any data. In the baseline I obtain $(\Pi_M, P_M)$ from the master solve through the aggregation identities (18) and (20) at the pinned wage $w_m = A_o$, whereas in the counterfactual exercises of Section 6 $(E_M, P_M)$ moves together with income through (34).

**Policy ($\alpha_p$, $\mu_p^{RVC}$, $\tau_p^{MFN}$).** The product-level RoO parameters $(\alpha_p, \mu_p^{RVC})$ come from USMCA Annex 4-B[7] and from Conconi et al.'s (2018) coding of NAFTA Annex 401, and

[7]Since Annex 4-B only exists as legal text, with no machine-readable version, I wrote a parser that goes

are refined with the INEGI 2018 input-output table in order to capture each product's input intensity. INEGI's I-O table is available at a six-digit SCIAN disaggregation, so I used Pierce and Schott (2012)'s HS-NAICS concordance tables to match the specific input content requirements determined at the HS system to the NAICS input requirements. More information on the matching between these two classifications is available in Appendix B. MFN tariffs $\tau_p^{MFN}$ come from the USITC 2023 Harmonized Tariff Schedule. RVC thresholds enter in their *net-cost* formulation rather than in their *transaction value* formulation (which is roughly 10pp higher), given that the condition (11) is a cost share. All of the parameters are attached at the HS6 level, which is the level at which Annex 4-B writes its product-specific rules, and also the level at which one is able to clearly distinguish between the three regimes. Codes from the HS2012, HS2017 and HS2022 revisions are all in the parameter file and are linked to each other through the UN Stats conversion tables, so that the customs codes of either vintage can find their rule. In both 2018 and 2023 the file gives a rule to 98% of Mexican manufacturing exports to the U.S., and what is left is Yarn-Forward.

$\alpha_p$ is computed from the INEGI input-output table conditional on the HS depth at which the rule is written, so that the resulting HS6 parameter still reflects the binding restriction. As a clarifying example, consider HS 8471.30 (portable automatic data processing machines, i.e. laptops). USMCA Annex 4-B specifies the following CTC restriction:

> *A change to a good of subheading 8471.30 from any other good within that subheading or any other subheading, except from subheading 8471.41 through 8471.50.*

The purpose of the carve-out is to close a loophole on substantial assembly. If it were not there, a Mexican firm could bring in a nearly complete processing-unit module under HS 8471.50 from China, fit it into a chassis, and the input would technically satisfy the tariff shift, given that 8471.50 is a different subheading from 8471.30. According to the input-output table, inputs in the 8471.41–50 range make up 14.6% of the intermediate spending of HS 8471.30, which in turn gives $\alpha_{8471.30} = 0.146$.[8]

Table 5 summarizes the resulting HS6 parameter universe that is used as a model input, and Figure 3 below shows the underlying trade-weighted distributions of $\alpha_p$ and $\mu_p^{RVC}$. On average across HS6 subheadings, pure-RVC coverage collapses to an automotive core at the

through the annex and records each product-specific rule (its CTC clauses, RVC thresholds and de minimis exceptions) in an HS6-level dataset. The treaty writes its rules with a handful of phrases that repeat over and over ("a change to X from any other Y", "a regional value content of not less than Z percent"), so the parser matches each rule to one of these templates in order to identify its regime and extract its thresholds. A rule that is written for a whole chapter or heading is passed down to all of its HS6 subheadings, except for those that have a rule of their own.

[8] In the parameter file that the estimation uses, this rule is coded as a change of subheading. The clause "any other good within that subheading" exempts the inputs classified in 8471.30 itself, but this coding restricts them as well, which adds another 0.3% of spending and gives $\alpha_{8471.30} = 0.149$.

75% net-cost threshold, Hybrid coverage roughly doubles, and the CTC input restriction $\alpha_p$ tightens on both its pure and its Hybrid arm.

Table 5: NAFTA vs USMCA Policy Parameters: Summary Stats

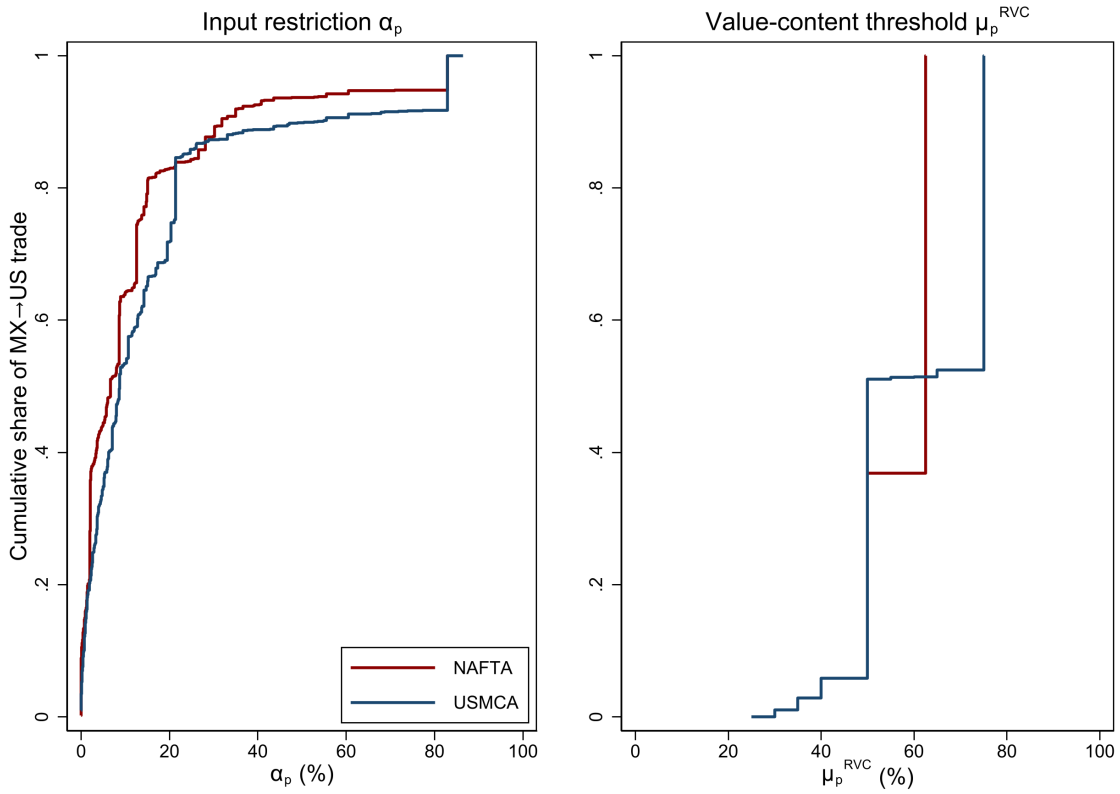
| | **NAFTA** | **USMCA** | **Δ** |
|---|---|---|---|
| *CTC regime* | | | |
| HS6 subheadings | 3,284 | 3,146 | −138 |
| $\alpha_p$ (mean / median) | 14.4 / 4.9 | 19.0 / 7.7 | +4.6 / +2.8 |
| *RVC regime* | | | |
| HS6 subheadings | 717 | 79 | −638 |
| $\mu_p^{RVC}$ (mean / median) | 50.0 / 50.0 | 71.9 / 75.0 | +21.9 / +25.0 |
| *Hybrid regime* | | | |
| HS6 subheadings | 818 | 1,575 | +757 |
| $\alpha_p$ (CTC arm, mean / median) | 4.5 / 0.4 | 23.6 / 16.3 | +19.1 / +15.9 |
| $\mu_p^{RVC}$ (RVC arm, mean / median) | 50.7 / 50.0 | 40.9 / 50.0 | −9.8 / 0.0 |
| *MFN tariff $\tau_p^{MFN}$ (common across policy regimes)* | | | |
| | | *Equal-wtd* | *TW (2023)* |
| $\tau_p^{MFN}$ (mean / median) | | 3.3 / 2.4 | 3.3 / 1.4 |
| Share with $\tau_p^{MFN} > 0$ | | 58.8% | 61.5% |
| Share with $\tau_p^{MFN} \geq 5$ | | 24.2% | 13.9% |

*Notes:* Values in %; Δ in pp. Equal-weighted across the 5,077 HS6 subheadings of the parameter file in chapters 16–97, which include codes of the HS2012, HS2017 and HS2022 revisions. TW (2023): weighted by 2023 U.S. imports from Mexico (UN Comtrade). RVC on net-cost basis.

**Closure ($\bar{L}$, $A_o$, $\chi$).** The labor-market closure adds a competitive non-manufacturing sector that pins the Mexican wage. Its productivity $A_o$ sets the level of that wage and equals the census wage bill per paid worker across manufacturing, trade and services, which amounts to 165.4 thousand 2023 pesos in the 2019 census and 168.7 thousand in 2024. The labor force $\bar{L}$ is the economically active population of the ENOE, averaged over the four quarters of 2018 and of 2023, 55.6 and 60.6 million people, so that the non-manufacturing sector stands for the rest of the economy. At that wage, manufacturing employs whatever labor its firms demand (17), and the competitive non-manufacturing sector employs everyone else.[9] Lastly, $\chi = 0.37$ is the share of Mexican final consumption that is spent on manufactures, which I take from the INEGI use table. It only weights the manufacturing price index in the welfare deflator (19) and plays no role in the demand loop (34).

[9]The counterfactual aggregates of Section 6 are evaluated at the firm choices of the executed solve, and the domestic demand loop (34) is solved again under this labor force. The firms are not re-optimized against the new demand, although this omission proves to be second order, as it stays below 0.02 percentage points of real income even at the largest surcharge. Intermediate purchases are assumed to scale with manufacturing sales.

Figure 3: Trade-weighted CDFs of the policy parameters $\alpha_p$ and $\mu_p^{RVC}$ under NAFTA and USMCA




*Notes:* 2018 weights for NAFTA, 2023 for USMCA.

**Compliance rate.** The compliance rate over all claims is the share of U.S. & CA-bound revenue that is shipped under preference, $\sum_{f,p} d_{fp} R_{f,p,\text{U.S. \& CA}} / \sum_{f,p} R_{f,p,\text{U.S. \& CA}}$, where $d_{fp}$ is the optimal choice in (14). Measured in this way, the model rate can be compared directly to the empirical benchmark in Table 3. Products that do not have a coded rule (1.2% of U.S. & CA-bound revenue in the 2018 panel and 1.8% in the 2023 panel) are never able to comply, and they do not pay a tariff either. Additionally, a zero-MFN product whose rule does not bind is able to comply at no cost at all, so the compliance rate that I report is the *strict* one, under which a product only counts as compliant if it faces a positive MFN rate. The data counterpart of this measure, the value claimed on positive-MFN lines over all value, is 52.3% in 2018 and 46.5% in 2024.

## 4.3 Structural Estimation

The remaining parameters are the firm-level sourcing-efficiency and market-reach vectors $\mathbf{a}_f = (a_{f,s})_{s \neq \text{U.S. \& CA}}$ and $\mathbf{b}_f = (b_{f,d})_{d \in \mathcal{D}}$, and the location parameters of the per-source and per-destination fixed-cost distributions $\{\mu_s, \mu_d\}$. The strategy consists in recovering $(\mathbf{a}_f, \mathbf{b}_f)$ for every firm by inverting the model on the cells where the firm is observed, and predicting them from a Heckman-corrected regression on the cells where it is not. I then estimate the fixed-cost location vector by inverting the model on aggregate participation shares, with one moment per parameter. The normalizing coordinate is $a_{f,\text{U.S. \& CA}} \equiv 1$, given that U.S. & CA is the source block with the highest participation (Table 7), meaning that the ratio denominator $\mu_{f,\text{U.S. \& CA}}$ is non-zero for the largest possible subsample of firms. No normalization is imposed on the demand side, as $b_{f,d}$ is inverted from (22) at all three destinations and its imputation regression is run on $\log b_{f,d}$ directly.

**Firm primitives ($\mathbf{a}_f$, $\mathbf{b}_f$): inversion on observed cells.** On the cells where firm $f$ actually sources from $s$ or sells to $d$, the primitives can be backed out directly from the model. Log-inverting the Fréchet share equation (7) at $a_{f,\text{U.S. \& CA}} \equiv 1$ gives the sourcing efficiency, while summing the CES revenue equation (13) across the firm's products at destination $d$ and solving for the market reach gives:

$$\log a_{f,s} = \tfrac{1}{\theta}\big[\log(\mu_{f,s}/\mu_{f,\text{U.S. \& CA}}) - \log(T_s(c_s\tau_s)^{-\theta})\big], \tag{21}$$

$$b_{f,d}^{\sigma-1} = \frac{\sum_p R_{f,p,d}}{E_d\, P_d^{\sigma-1} \sum_{p \in \mathcal{P}_f} \varpi_{f,p}\, p_{f,p,d}(\mathcal{S}_f, d_{fp})^{\,1-\sigma}}, \tag{22}$$

where $(E_d, P_d)$ is the market bundle of destination $d$, Mexico included, set as in Section 4. These inversions are exact on $\{f : \mu_{f,s} > 0,\ \mu_{f,\text{U.S. \& CA}} > 0\}$ and $\{f : Y_{f,d} > 0\}$, and are undefined elsewhere. Filling the unsourced cells with zeros would eliminate the extensive margin in the counterfactuals, given that no firm could ever flip into a new sourcing country or destination under a policy shock. I instead fill the missing cells with a Heckman-corrected regression prediction.

**Firm primitives ($\mathbf{a}_f$, $\mathbf{b}_f$): Heckman two-step on missing cells.** If one were to estimate the mean functions in (21) and (22) through standard OLS on the selected subsample, the estimates would be biased upward. The reason behind this is that the firms that are observed sourcing from $s$ tend to be the ones with a high latent $a_{f,s}$, so the fitted coefficients would overstate the unconditional expectation. In order to deal with this issue, I stack the firm–source pairs of the five non-U.S. & CA source blocks into a single long panel (U.S. & CA is the normalizing coordinate and is therefore not imputed) and fit a single Heckman (1979) two-step regression for the latent sourcing efficiency, and I follow by fitting the analogous

regression for the latent market reach, stacking all three destinations. Each of these regressions also pools the two vintages. Specifically, the 2018 and 2023 panels are stacked with a vintage fixed effect in both stages, meaning that the two equilibria share the same imputation surface up to a level shift and that the imputed cells of each vintage do not rest only on the sample of that vintage. The sourcing panel is made up of the five blocks times the firms with $\mu_{f,\text{U.S. \& CA}} > 0$, which are the firms for which the normalized log-ratio is defined (4,939 in 2018 and 4,452 in 2023), for a total of 46,955 firm–block rows, and the remaining 780 and 681 firms receive fitted values. The demand panel is made up of the three destinations times the exporters of both vintages, meaning the firms with U.S. & CA or RoW sales (4,921 and 4,449), for a total of 28,110 rows. I do not include the import-only firms in this panel, given that they have no foreign sales by construction and the selection equation would treat them as firms that chose to stay out of both foreign markets. They receive fitted values on the foreign coordinates instead. A similar correction is used by Helpman et al. (2008) for the selection of firms into export markets, and in this paper I apply it to the selection into source markets as well.

$$\text{Stage 1 (selection):} \quad \Pr(\mu_{f,s} > 0 \mid X_{f,s}, W_f) = \Phi(X'_{f,s}\delta_a + \delta_a^W W_f), \tag{23}$$

$$\text{Stage 2 (outcome):} \quad \log(\mu_{f,s}/\mu_{f,\text{U.S. \& CA}}) = X'_{f,s}\beta_a + \lambda_a \,\text{IMR}(X'_{f,s}\hat{\delta}_a + \hat{\delta}_a^W W_f) + u_{f,s}\ , \tag{24}$$

where the regressor vector of the outcome stage $X_{f,s}$ includes firm size ($\log L_f$), source-block fixed effects, ownership-country fixed effects, NAICS-3 sector fixed effects and the vintage fixed effect, whereas $W_f$ is the *exclusion restriction*, which only enters the selection stage. The exclusion is the number of establishments of the firm ($\log N_{estab}$) on the sourcing side and the breadth of its product range ($\log N_{HS6}$) on the demand side, and the first-stage coefficient on each of them is reported in Table 8, while $\text{IMR}(z) = \phi(z)/\Phi(z)$ denotes the inverse Mills ratio. Note that the differences across sources are captured by the source fixed effects, as all of the blocks are pooled into one regression and these fixed effects absorb the mean gap in log shares between them.

The choice of exclusions is based on the distinction that the model makes between the extensive and the intensive margin. I use the number of establishments on the sourcing side under the notion that a firm with several plants may find it easier to pay the fixed cost of a relationship with a foreign supplier (procurement staff, customs registration, supplier search), as it can spread this cost over a larger operation. This can make the firm more likely to source from a given country, but once it sources from $s$ the share equation (7) is

scale-free and only depends on $a_{f,s}$. Similarly, I use the breadth of the product range on the demand side under the notion that a firm with more products obtains a larger variable profit from paying the entry cost of a destination once, whereas for a firm that already serves $d$ the inversion (22) nets the product mix out of $b_{f,d}$. Thus, breadth should not move market reach itself. The identifying assumption is that, conditional on the included controls, the number of establishments and product breadth only shift the capacity to pay fixed costs, and are not proxies for latent sourcing efficiency or market reach. If $(\varepsilon_{f,s}, u_{f,s})$ is bivariate normal, the linear-IMR form is the closed-form correction, with $\lambda_a = \rho_a \sigma_u$, and the fitted $\hat{\beta}_a$ then recovers the bias-corrected unconditional mean $X'_{f,s}\hat{\beta}_a = \mathbb{E}[\log(\mu_{f,s}/\mu_{f,\text{U.S. \& CA}}) \mid X_{f,s}]$. Plugging this into (21) gives the imputed $\hat{a}_{f,s}$ on the missing cells:

$$\log \hat{a}_{f,s} = \tfrac{1}{\theta}\left[X'_{f,s}\hat{\beta}_a - \log(T_s(c_s\tau_s)^{-\theta})\right] \quad \text{for } \mu_{f,s} = 0 \text{ or } \mu_{f,\text{U.S. \& CA}} = 0,$$

whereas the closed-form inversion (21) is kept for the observed cells. Following the same procedure on the pooled firm–destination panel gives $\hat{b}_{f,d}$. Imputed cells are capped, on both sides, at the largest value inverted from data in the same block or destination, and the resulting firm-by-source and firm-by-destination matrices, $\hat{\mathbf{A}}$ and $\hat{\mathbf{B}}$, are what allows firms to enter new sources and destinations in the counterfactuals, since an inactive cell whose imputed value is high enough to clear the fixed cost can become active after a policy shock.

**Fixed costs ($\kappa_s$, $\kappa_d$): moment inversion on aggregate participation shares with lognormal draws.** Firm-country and firm-destination fixed costs are drawn iid from log-normal distributions, following Antràs et al. (2017) and Morán and Cebreros (2025):

$$\kappa_{f,s} \sim \text{LogNormal}(\mu_s,\, \delta), \qquad \kappa_{f,d} \sim \text{LogNormal}(\mu_d,\, \delta), \tag{25}$$

with one draw per firm per source country and one per firm per destination. $\kappa$ is the fixed cost that enters the firm's profit function (14), and $\mu$ is the log-space location parameter of its distribution, with $E[\kappa_c] = \exp(\mu_c + \delta^2/2)$. I pin the dispersion at $\delta = \sqrt{\log 3}$, following Morán and Cebreros (2025).

Following Antràs et al. (2017), I pin $\kappa_s^{MX} \equiv 0$, given that every Mexican manufacturer sources domestically by construction. Even if a firm sources no material inputs domestically, every firm sources at least some labor domestically, so there is no meaningful *entry* fixed cost into the Mexican input market. The $\mu$ of the remaining eight blocks (five non-MX sources and three destinations) are estimated. The estimation recovers the eight location parameters

$$\boldsymbol{\mu} = \left\{\mu_s^{\text{U.S. \& CA}},\, \mu_s^{AX},\, \mu_s^{CH},\, \mu_s^{EU},\, \mu_s^{RoW},\, \mu_d^{\text{U.S. \& CA}},\, \mu_d^{MX},\, \mu_d^{RoW}\right\},$$

one per estimated block. The results table reports $\hat{\kappa}_c = \exp(\hat{\mu}_c + \delta^2/2)$, and the rest of the paper refers to fixed costs as $\kappa$.

*Model solution for a given $\boldsymbol{\mu}$.* For a given $\boldsymbol{\mu}$, every firm has its own draw $\kappa_{f,c} = \exp(\mu_c + \delta\, z_{f,c})$ in each of the nine blocks ($\kappa_{f,s}^{MX} = 0$ is pinned), where the standard normals $z_{f,c}$ are drawn once and kept fixed across evaluations so that the simulated moments use common random numbers. Solving (14) once per firm at these draws then gives each firm's sourcing set $\mathcal{S}_f$ and destination set $\mathcal{D}_f$.

*Moment vector.* Each of the eight location parameters is matched to one participation moment, with the five non-MX source-side fixed costs identified from the extensive margin of sourcing,

$$m_s^{\text{source}} = \Pr\big(\mu_{f,s} > 0\big), \quad s \in \{\text{U.S. \& CA},\ AX,\ CH,\ EU,\ RoW\}.$$

The other three identify the destination-side fixed costs through the extensive margin of selling into each destination:

$$m_d^{\text{dest}} = \Pr\big(Y_{f,d} > 0\big), \quad d \in \{\text{U.S. \& CA},\ MX,\ RoW\}.$$

Additional moments are computed but not targeted, and I use them as checks in Section 5.1. They include the aggregate compliance rate (the share of MX→US import value claimed under preference, from USA Trade Online) and the share of firms sourcing from at least two foreign blocks, which depends on how participation is correlated across blocks and is therefore not pinned down by the marginal shares.

*Estimator.* With $\hat{\boldsymbol{m}}(\boldsymbol{\mu})$ the eight simulated participation rates at $\boldsymbol{\mu}$, $\boldsymbol{m}$ their data counterparts and $\hat{\boldsymbol{g}}(\boldsymbol{\mu}) = \hat{\boldsymbol{m}}(\boldsymbol{\mu}) - \boldsymbol{m} \in \mathbb{R}^8$ the gaps between them, the estimator is the root

$$\hat{\boldsymbol{\mu}}: \quad \hat{\boldsymbol{g}}(\hat{\boldsymbol{\mu}}) = \mathbf{0}. \tag{26}$$

Participation in block $c$ is weakly decreasing in $\mu_c$, given that a higher $\mu_c$ raises $\kappa_{f,c}$ for every firm, which can only take firms out of the block, meaning that each gap is monotone in its own parameter. Having eight parameters and eight monotone moments, (26) is exactly identified, which allows one to solve it by inversion instead of minimizing a distance. This corresponds to the just-identified case of the simulated method of moments, where the moment conditions (26) hold with equality up to the $1/N$ step of the simulated shares. Recovering parameters through an exact inversion, with one moment per parameter, is common practice in quantitative trade models. To state an example, Melitz and Redding (2015) choose the fixed exporting cost so as to reproduce the observed share of exporting firms, and, as Redding and

Rossi-Hansberg (2017) note for the class of exactly identified models that grew out of Eaton and Kortum (2002), observed outcomes map one-to-one into the unobserved primitives that rationalize them as an equilibrium.

# 5 Estimation Results

Section 5.1 reports the structural estimates and the fit of the model to targeted and untargeted moments in both vintages. Section 5.2 then examines the estimated firm primitives in the 2023 vintage, while Appendix E reports the same objects for 2018.

## 5.1 Parameter Estimates and Model Fit

Table 6: Structural parameter estimates, both vintages

| | 2018 vintage | | 2023 vintage | |
|---|---|---|---|---|
| Parameter | $\hat{\kappa}$ | $\text{SE}_\kappa$ | $\hat{\kappa}$ | $\text{SE}_\kappa$ |
| *Sourcing fixed costs* $\kappa_s$ | | | | |
| U.S. & CA | 3.68 | 0.42 | 3.52 | 0.47 |
| Asia excl. China | 0.92 | 0.07 | 0.98 | 0.09 |
| China | 0.67 | 0.07 | 0.80 | 0.09 |
| Europe | 0.76 | 0.08 | 1.04 | 0.09 |
| Rest of World | 0.42 | 0.04 | 0.60 | 0.05 |
| *Destination entry fixed costs* $\kappa_d$ | | | | |
| U.S. & CA | 15.40 | 1.26 | 19.90 | 1.65 |
| Mexico | 49.68 | 4.45 | 50.95 | 5.95 |
| Rest of World | 4.52 | 0.65 | 4.76 | 0.74 |
| Firms | 5,719 | | 5,133 | |
| firm×HS6 products | 136,303 | | 122,478 | |

*Notes:* $\hat{\kappa}$ is the mean of the fitted distribution, $\exp(\hat{\mu}+\delta^2/2)$. $\text{SE}_\kappa$ is the standard deviation of $\hat{\kappa}$ (not of $\hat{\mu}$) across $B = 100$ firm-cluster bootstrap replicates. Fixed costs are measured in Mexican labor, as worker-years at the pinned wage, and $\kappa_s^{\text{MX}} \equiv 0$.

Table 6 reports the estimates, and two things about them are worth pointing out. The first has to do with the size of the fixed costs. Sourcing from U.S. & CA shows to be the most expensive option in both vintages, at about 3.5 worker-years, while no other source block costs more than about one worker-year. Between the two vintages the U.S. & CA cost falls by 4% and all the others rise, by +6% for Asia excluding China, +19% for China, +36% for Europe and +42% for the Rest of World. On the destination side the domestic market is by far the most expensive, at about 50 worker-years in both vintages. Almost every firm has a high inverted domestic market reach, and a large entry cost is the only way to square that

with the 11 to 14% of firms that do not sell at home. The U.S. & CA entry cost rises by 29% and the Rest-of-World cost by 5%. In terms of revenue these fixed costs are modest, with domestic entry, the largest, at about 0.3% of mean firm revenue and U.S. & CA sourcing at about 0.02%. Second, the eight location parameters are all interior, with bootstrap intervals well inside the inversion bracket, and the inversion hits every participation target (Table 7).

**Model fit.** The inversion matches all eight targeted participation shares (Table 7). Among the untargeted moments, the share of firms sourcing from at least two foreign blocks is close to the data (88.7% against 86.6% in 2018, 90.5% against 87.4% in 2023), and so are the U.S. & CA share of revenue (44% and 42%) and total sales, within 4% of the census. The model overpredicts the compliance rate (59.0% against 52.3% in 2018, 60.7% against 46.5% in 2023). Over all claims, where zero-MFN products comply at no cost, the gap widens to 86.5% against 56% and 86.8% against 48%. This gap is likely due to a cost of claiming preference which the model does not include. One can observe in the data that, once the 2025 surcharge made non-compliance expensive, compliance went from 44% to 85% (Table 3). Thus, the counterfactuals speak more to the response to policy than to the level of utilization at low tariffs.

**Selection-corrected imputation.** The outcome stage of the two pooled Heckman regressions is in Table 8. On both sides the inverse-Mills loading shows to be positive and precisely estimated, at 3.76 (s.e. 0.36) for sourcing efficiency and 2.62 (s.e. 0.30) for market reach. Meaning that the firms that are observed sourcing from a block, or selling in a market, have a higher latent sourcing efficiency or market reach than what their observables would predict. Unadjusted OLS on the observed cells would therefore overstate the latent means, and the unconditional fit that fills in the missing cells lies below the observed cells. Regarding the sourcing-efficiency surface, which is estimated on the log share ratio, that is, $\theta \log a_{f,s}$ up to a block constant (21), it rises with firm size and shifts up by 0.19 in the 2023 vintage (0.11 in $\log a_{f,s}$).

**Estimated firm primitives.** Table 9 shows the distribution across firms of $\log a_{f,s}$ and $\log b_{f,d}$ that comes out of the inversions and the Heckman-corrected fits described above, along with the share of firms for which each coordinate is inverted from data instead of imputed. Sourcing efficiencies are widely spread (their standard deviations go from 1.5 to 1.6 log units) and follow a clear ordering across sources, with Mexico having the highest mean, the AX, CH and EU blocks grouped around a common lower mean and RoW below them. The demand coordinates are ordered in the same way as in the data, with Mexico first and the Rest of World, a market that just over half of the firms serve, three to four log units below the other two.

Table 7: Targeted and untargeted moments: data vs. model, both vintages

| | 2018 vintage | | | 2023 vintage | | |
|---|---|---|---|---|---|---|
| Moment | Data | Model | Diff | Data | Model | Diff |
| *Source-side participation shares* $\Pr(\mu_{f,s} > 0)$ *(targeted)* | | | | | | |
| $s =$ U.S. & CA | 0.864 | 0.864 | +0.000 | 0.867 | 0.868 | −0.000 |
| $s =$ AX | 0.738 | 0.739 | −0.000 | 0.762 | 0.762 | −0.000 |
| $s =$ CH | 0.784 | 0.784 | +0.000 | 0.815 | 0.815 | +0.000 |
| $s =$ EU | 0.759 | 0.759 | −0.000 | 0.770 | 0.770 | +0.000 |
| $s =$ RoW | 0.646 | 0.645 | +0.000 | 0.650 | 0.650 | +0.000 |
| *Destination-side participation shares* $\Pr(Y_{f,d} > 0)$ *(targeted)* | | | | | | |
| $d =$ U.S. & CA | 0.780 | 0.780 | −0.000 | 0.799 | 0.799 | −0.000 |
| $d =$ MX | 0.861 | 0.861 | +0.000 | 0.892 | 0.892 | +0.000 |
| $d =$ RoW | 0.527 | 0.527 | −0.000 | 0.555 | 0.556 | −0.000 |
| *Compliance (untargeted)* | | | | | | |
| Compliance rate | 0.523 | 0.590 | −0.067 | 0.465 | 0.607 | −0.142 |
| *Revenue share by destination (untargeted)* | | | | | | |
| U.S. & CA | 0.444 | 0.440 | +0.004 | 0.416 | 0.415 | +0.001 |
| Mexico | 0.494 | 0.499 | −0.005 | 0.529 | 0.531 | −0.001 |
| RoW | 0.062 | 0.061 | +0.000 | 0.054 | 0.054 | +0.000 |
| *Share of materials by source (untargeted)* | | | | | | |
| U.S. & CA | 0.328 | 0.294 | +0.034 | 0.334 | 0.255 | +0.080 |
| Mexico | 0.367 | 0.432 | −0.065 | 0.328 | 0.466 | −0.139 |
| Asia excl. China | 0.116 | 0.104 | +0.012 | 0.122 | 0.099 | +0.023 |
| China | 0.096 | 0.085 | +0.011 | 0.102 | 0.084 | +0.018 |
| Europe | 0.061 | 0.056 | +0.005 | 0.070 | 0.059 | +0.011 |
| Rest of World | 0.032 | 0.030 | +0.002 | 0.044 | 0.038 | +0.006 |

*Notes:* Diff = Data − Model. Compliance rate: under NAFTA rules in 2018 and USMCA rules in 2023 (Section 4.2). Revenue and material shares: means of firm shares weighted by census revenue and census materials, at each vintage's baseline solve (2018 under USMCA rules).

## 5.2 Validating the Estimated Firm Primitives

This subsection reports the 2023 vintage throughout, while Appendix E reports each object for the 2018 vintage.

**Ownership and the geography of the primitives.** Tables 10a and 10b split the two blocks of Table 9 by the ownership country of the firm for the 2023 vintage. Perhaps the most interesting observation on the sourcing side is that the geography lines up with ownership. Whenever the home region of a group is part of the sourcing menu, that is the group's best coordinate (the bold diagonal of Table 10a). Mexican-owned firms have their highest sourcing efficiency in Mexico (+0.86), AX-owned firms in Asia excluding China (+1.52, which is the largest entry in the panel), Chinese-owned firms in China (+1.40) and

Table 8: Heckman second stage: sourcing-efficiency and market-reach imputation equations

| | Sourcing efficiency $\log(\mu_{f,s}/\mu_{f,\text{U.S. \& CA}})$ | Market reach $\log b_{f,d}$ |
|---|---|---|
| $\log L_f$ | 0.329 | 0.400 |
| | (0.034) | (0.018) |
| Inverse Mills ratio ($\hat{\lambda}$) | 3.756 | 2.620 |
| | (0.362) | (0.296) |
| Constant | −2.523 | 6.366 |
| | (0.456) | (0.152) |
| *First stage (selection probit)* | | |
| Exclusion: $\log N_{estab}$ | 0.043 | |
| | (0.011) | |
| Exclusion: $\log N_{HS6}$ | | 0.086 |
| | | (0.008) |
| Source FE | Yes | |
| Destination FE | | Yes |
| Ownership FE | Yes | Yes |
| Sector (NAICS-3) FE | Yes | Yes |
| Year FE | Yes | Yes |
| $N$ (panel) | 46,955 | 28,110 |
| $N$ (selected) | 36,370 | 22,466 |

*Notes:* Heckman (1979) two-step estimated on the 2018 and 2023 panels stacked together and pooled across blocks, with a vintage fixed effect in both stages. Standard errors in parentheses. The ownership fixed effects take Mexico as the base, with six groups on the sourcing side (the source blocks) and three on the demand side (Mexico, U.S. & CA and other foreign). Each exclusion only enters the selection probit of its own side. The sourcing-efficiency panel consists of the firms with $\mu_{f,\text{U.S. \& CA}} > 0$ times five blocks, and the demand panel of the exporters (firms with U.S. & CA or RoW sales) times three destinations.

EU-owned firms in Europe (+0.74). For U.S. & CA-owned firms the home block is the normalizing one ($a_{f,\text{U.S. \& CA}} \equiv 1$), and they lie below zero on every other block, so for them too the home block is the best. Notably, the peaks of the AX- and CH-owned firms are about two log points above the means of the Mexican-owned firms in those same blocks. The RoW-owned group, which pools several origins, is the only one that does not peak in its own block. On the demand side the pattern repeats itself (Table 10b), as the group with the highest mean market reach in each market is the one from that region, with Mexican-owned firms in Mexico (10.3), U.S. & CA-owned firms in the U.S. & CA market (9.5) and the remaining foreign owners in the Rest of World (6.2).

These differences in levels have a direct effect on what firms do. Through the lens of the model, a higher $a_{f,s}$ makes a firm more likely to pay the fixed cost of block $s$ and, once it does, to spend a larger share $\mu_{f,s}$ on that block, while a higher $b_{f,d}$ raises the revenue of the firm in market $d$. Foreign affiliates thus source more from the home regions of their parents and sell more there as well. Note that this pattern comes out of the inverted cells.

Table 9: Estimated firm primitives: cross-firm distribution (2023 vintage)

| Coordinate | mean | sd | observed |
|---|---|---|---|
| *Sourcing efficiency* $\log a_{f,s}$ *(U.S. & CA* $\equiv 0$*)* | | | |
| Mexico | 0.377 | 1.480 | 0.55 |
| Asia excl. China | −0.581 | 1.532 | 0.73 |
| China | −0.502 | 1.575 | 0.75 |
| Europe | −0.451 | 1.519 | 0.72 |
| Rest of World | −0.923 | 1.597 | 0.62 |
| *Market reach* $\log b_{f,d}$ | | | |
| U.S. & CA | 8.583 | 2.185 | 0.80 |
| Mexico | 9.828 | 2.163 | 0.89 |
| Rest of World | 5.519 | 2.353 | 0.56 |

*Notes:* 5,133 traders. Direct inversion on observed cells; Heckman-fitted mean on missing cells, capped at the largest inverted value of the coordinate. "observed": share of firms with the coordinate inverted from data; a sourcing-efficiency coordinate counts as observed only for firms that also source from U.S. & CA, the normalizing block. $\log a_{f,\text{U.S. \& CA}} \equiv 0$. Means and standard deviations aggregate the ownership-group statistics of the replication workbook by firm counts. 2018 vintage in Appendix E.

Table 10a: Estimated sourcing efficiency by ownership group (2023 vintage)

| | Ownership group | | | | | |
|---|---|---|---|---|---|---|
| | Mexico | U.S. & CA | Asia excl. China | China | Europe | Rest of World |
| Firms | 2,639 | 1,470 | 228 | 23 | 513 | 260 |
| *Mean sourcing efficiency* $\log a_{f,s}$*, by source block (U.S. & CA* $\equiv 0$*)* | | | | | | |
| Mexico | **0.86** | −0.61 | 0.85 | 0.57 | 0.55 | 0.35 |
| U.S. & CA | 0.00 | **0.00** | 0.00 | 0.00 | 0.00 | 0.00 |
| Asia excl. China | −0.74 | −0.71 | **1.52** | 0.58 | −0.45 | −0.50 |
| China | −0.50 | −0.70 | 0.09 | **1.40** | −0.25 | −0.52 |
| Europe | −0.36 | −1.11 | −0.52 | −0.47 | **0.74** | 0.03 |
| Rest of World | −0.82 | −1.46 | −0.94 | −0.55 | −0.15 | **−0.44** |

*Notes:* Means over all firms in each group, observed and Heckman-imputed cells together; per-coordinate observed shares in Table 9. Bold: each group's home block. $\log a_{f,\text{U.S. \& CA}} \equiv 0$ for every firm, so the U.S. & CA row is zero and each entry is a sourcing efficiency relative to U.S. & CA sourcing. Owners from Asia excl. China are those from Japan, Korea and Singapore, the Asian origins coded in both censuses; owners from other Asian countries are in the Rest of World group. 2018 vintage in Appendix E.

**Variation explained by observables.** Table 11 reports the share of the cross-firm variation in each $\log a_{f,s}$ and $\log b_{f,d}$ latent that is explained by ownership-country and industry (NAICS-3) fixed effects and their interaction. Let the primitive coordinates be indexed by $k$, which is a source block $s$ on the sourcing side or a destination $d$ on the demand side, and let $x_{f,k}$ be firm $f$'s estimated value of coordinate $k$, that is, $x_{f,k} = \log a_{f,s}$ for $k = s$ and $x_{f,k} = \log b_{f,d}$ for $k = d$, inverted or imputed as in Table 9. Let $o(f)$ be the firm's ownership

Table 10b: Estimated market reach by ownership group (2023 vintage)

| | Ownership group | | |
|---|---|---|---|
| | Mexico | U.S. & CA | Other foreign |
| Firms | 2,639 | 1,470 | 1,024 |
| *Mean market reach* $\log b_{f,d}$*, by destination* | | | |
| Mexico | **10.34** | 9.06 | 9.61 |
| U.S. & CA | 8.12 | **9.48** | 8.49 |
| Rest of World | 5.37 | 5.32 | **6.19** |

*Notes:* Means over all firms in each group, observed and Heckman-imputed cells together. Owners are grouped by destination block, so "Other foreign" pools the firms owned from Asia excl. China, China, Europe and the Rest of World in Table 10a. Bold: each group's home market. 2018 vintage in Appendix E.

group and $n(f)$ its NAICS-3 sector. For each coordinate, I estimate four regressions by OLS,

$$
\begin{aligned}
x_{f,k} &= \xi^{o}_{k,o(f)} + e^{\text{own}}_{f,k}, & x_{f,k} &= \xi^{n}_{k,n(f)} + e^{\text{ind}}_{f,k}, \\
x_{f,k} &= \xi^{o}_{k,o(f)} + \xi^{n}_{k,n(f)} + e^{\text{both}}_{f,k}, & x_{f,k} &= \xi^{on}_{k,o(f),n(f)} + e^{\times}_{f,k},
\end{aligned}
\tag{27}
$$

where $\xi^{o}_{k,o}$ is a fixed effect for ownership group $o$, $\xi^{n}_{k,n}$ is a fixed effect for sector $n$, $\xi^{on}_{k,o,n}$ is a fixed effect for each ownership–sector cell, all of them specific to coordinate $k$, and $e^{m}_{f,k}$ is the residual of specification $m$. The table reports the $R^2$ of each of these regressions. The goal of this exercise is to address a natural question regarding the Heckman imputation, which is how much of the heterogeneity in $(\mathbf{a}_f, \mathbf{b}_f)$ is picked up by the coarse observables (ownership and sector) that go into the mean function, and how much of it is idiosyncratic. Even when they are fully interacted, the observables explain at most 25% of the variation across firms in any of the primitive coordinates. Ownership shows to be the stronger of the two for the MX, EU and RoW sourcing-efficiency coordinates and ties with sector for AX, whereas sector is the stronger one for China and for all three demand coordinates. It is important to mention that the $R^2$ are biased upward, given that the imputed cells are fitted values of a mean function that contains these same ownership and sector dummies. Even so, the conclusion that observables account for a minority of the variation holds in spite of this bias.

**Cross-country correlation structure of the primitives.** Tables 12a and 12b report the sample correlation of $(\log a_f, \log b_f)$ across the 2023 firms. An interesting observation in the sourcing-efficiency block (Table 12a) is that there is a clear East-Asian cluster, given that the AX–CH correlation of 0.54 is the largest pairwise correlation, above the 0.26 to 0.49 range observed for the other pairs of blocks. Firms with an above-average sourcing efficiency in China tend to have an above-average sourcing efficiency in AX as well, which is consistent with a latent "regional supplier-network access" factor. This correlation helps explain part of

Table 11: Share of the cross-firm variation in $\log a_{f,s}$ and $\log b_{f,d}$ explained by ownership and sector (2023 vintage)

| | | $R^2$ from fixed effects for | | | |
|---|---|---|---|---|---|
| Latent | Coordinate | Own | Sector | Own + Sector | Own × Sector |
| $\log a$ | MX | 0.188 | 0.093 | 0.234 | 0.254 |
| | AX | 0.094 | 0.094 | 0.173 | 0.189 |
| | CH | 0.020 | 0.069 | 0.094 | 0.114 |
| | EU | 0.122 | 0.045 | 0.158 | 0.177 |
| | RoW | 0.063 | 0.045 | 0.099 | 0.119 |
| $\log b$ | U.S. & CA | 0.071 | 0.186 | 0.223 | 0.237 |
| | MX | 0.068 | 0.103 | 0.151 | 0.163 |
| | RoW | 0.021 | 0.111 | 0.138 | 0.151 |

*Notes:* 5,133 traders. $R^2$ from OLS on ownership dummies alone, NAICS-3 dummies alone, their union, and full interaction, equation (27). Ownership has six levels for the sourcing-efficiency rows and three levels (Mexico, U.S. & CA, other foreign) for the market-reach rows. 2018 vintage in Appendix E.

the sourcing-reallocation pattern in Section 6.3, where sourcing shifts toward Asia excluding China about twice as much, in proportion, as toward any other block when China becomes more expensive. The demand block (Table 12b) shows weak and positive correlations, at 0.29 for U.S. & CA–RoW, 0.26 for Mexico–RoW and 0.13 for U.S. & CA–Mexico, meaning that firms with a high market reach in one foreign market tend to have it in the other one as well, rather than specializing in one of them.

Table 12a: Correlation structure of estimated sourcing efficiency $a_{f,s}$ (2023 vintage)

| | MX | AX | CH | EU | RoW |
|---|---|---|---|---|---|
| MX | 1.00 | | | | |
| AX | .26 | 1.00 | | | |
| CH | .33 | .54 | 1.00 | | |
| EU | .47 | .35 | .38 | 1.00 | |
| RoW | .38 | .30 | .32 | .49 | 1.00 |

*Notes:* 5,133 traders. Lower triangle of the correlation matrix (upper triangle omitted by symmetry). U.S. & CA is omitted ($\log a_{f,\text{U.S. \& CA}} \equiv 0$). 2018 vintage in Appendix E.

# 6 Counterfactual Analysis

This section begins by sweeping individual policy instruments around the 2023 baseline, which are the rules of origin (Section 6.1), the MFN tariff (Section 6.2), tariffs on Chinese inputs (Section 6.3), the interaction between content rules and the outside tariff (Section 6.4)

Table 12b: Correlation structure of estimated market reach $b_{f,d}$ (2023 vintage)

| | U.S. & CA | MX | RoW |
|---|---|---|---|
| U.S. & CA | 1.00 | | |
| MX | .13 | 1.00 | |
| RoW | .29 | .26 | 1.00 |

*Notes:* 5,133 traders. Lower triangle of the correlation matrix (upper triangle omitted by symmetry). 2018 vintage in Appendix E.

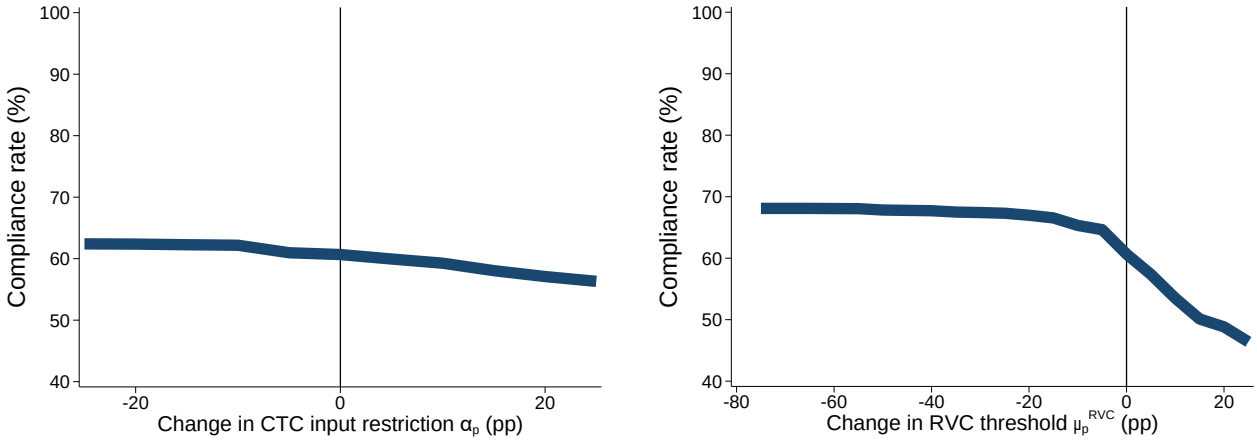
and the agreement itself (Section 6.5), and it ends with a decomposition of the observed 2018–2023 transition (Section 6.6). It is worth pointing out that, unlike the decomposition, each of these policy experiments changes the cost of the option that was the cheapest for a firm at baseline, given that a tighter rule of origin raises the cost of complying, whereas a higher tariff raises the cost of not complying. Each firm then re-optimizes, meaning that responses are heterogeneous: for some firms not complying remains the cheapest option, for others complying becomes the cheapest option, and some others keep complying at a higher cost. The aggregate patterns reported below summarize these firm-level choices. In every counterfactual, the foreign pairs $(E_d, P_d)$ are held fixed and the Mexican pair $(E_M, P_M)$ moves with income through the demand loop (34) (Section 3.4).

## 6.1 Rules-of-Origin Strictness

Figure 4 shows what happens when the two content parameters are swept around their USMCA baselines while everything else is held fixed, with the CTC input restriction $\alpha_p$ moving over ±25pp and the RVC threshold $\mu_p^{RVC}$ over $[-75, +25]$pp. One can observe that compliance responds strongly to both sweeps and that the response differs across regimes.

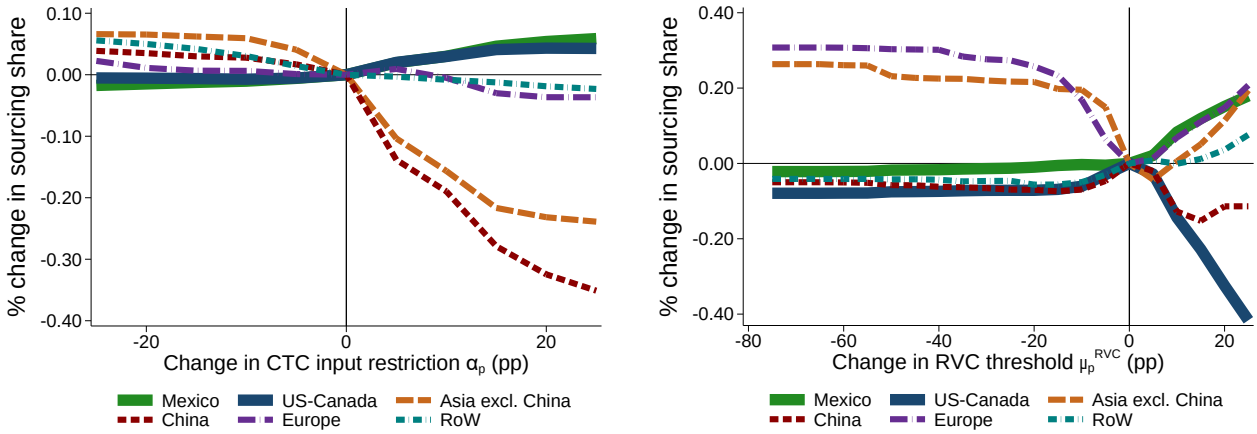
There are two main takeaways from this figure. The first one is the size of the compliance response. A ±25pp swing in $\alpha_p$ around the USMCA baseline moves compliance by +1.7pp at the loosened end and by −4.4pp at the tightened end, a range of 6pp. This response comes mainly from the CTC regime, whose rate falls from 98% to 43% over the sweep, and from the Hybrid regime, which gives up 7pp relative to the baseline at the tight end even though it is able to fall back on its RVC arm. The $\mu_p^{RVC}$ sweep over $[-75, +25]$pp gives a 22pp range (+7.4pp at the loose end and −14.3pp at the tight end). Given that RVC compliance is priced rather than binary (Appendix A.4), the response to a tighter threshold is a smooth one. Firms whose content falls short buy the North American share that they are missing through the restriction $\alpha^*$, up to the point where its cost exceeds the tariff, so that the RVC rate declines gradually from 81% at the baseline to 42% at +25pp instead of collapsing at a cutoff, and it reaches 99% once the threshold is loosened by 40pp or more. Note that

Figure 4: Counterfactual sweeps of rules-of-origin strictness



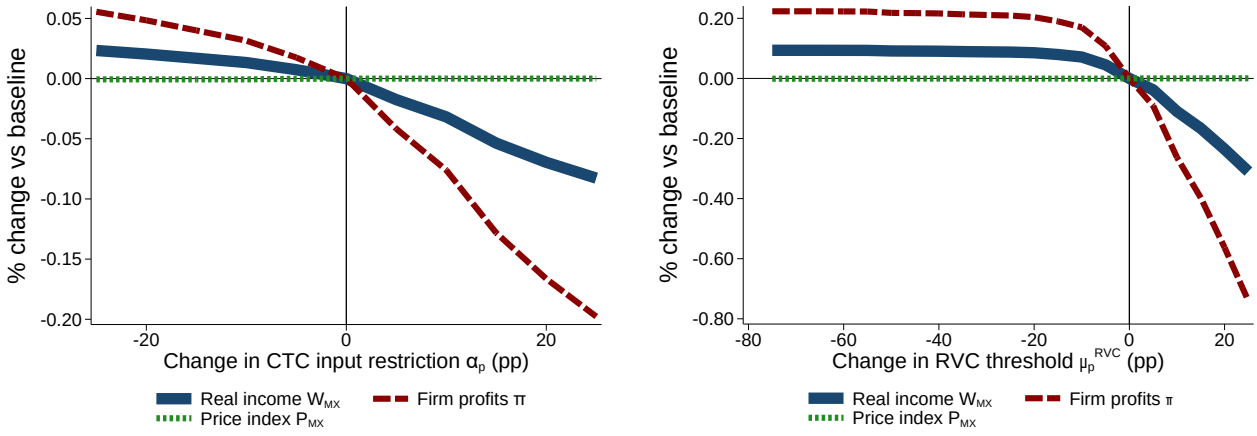

(a) Compliance ($\alpha_p$ sweep)

(b) Compliance ($\mu_p^{RVC}$ sweep)




(c) Sourcing composition ($\alpha_p$ sweep)

(d) Sourcing composition ($\mu_p^{RVC}$ sweep)




(e) Welfare and components ($\alpha_p$ sweep)

(f) Welfare and components ($\mu_p^{RVC}$ sweep)

*Notes:* 2023 vintage. Horizontal axis: change in pp from the USMCA value of $\alpha_p$ (left column) or $\mu_p^{RVC}$ (right column); the vertical line marks the baseline. Top row: compliance rate of Section 4.2 (positive-MFN lines). Middle row: % change relative to the baseline in each origin's share of Mexican sourcing, $\bar{\mu}_s = M_{MX,s} / \sum_{s'} M_{MX,s'}$. Bottom row: % change in real income, the traders' profits (non-traders excluded) and the Mexican price index. Given that the price index moves by less than 0.001% along these sweeps, its line coincides with zero.

the response is regime-specific by construction, as only CTC and Hybrid move under the $\alpha_p$ sweep, whereas only RVC and Hybrid move under the $\mu_p^{RVC}$ sweep.

The second is that the welfare panels (e, f) are not flat. Tightening $\alpha_p$ by 25pp lowers $W_M$ by 0.08% and the traders' profits by 0.20%, whereas tightening $\mu_p^{RVC}$ by 25pp lowers them by 0.31% and 0.74%, and loosening raises $W_M$ by up to 0.02% and 0.09% respectively. The price index moves by less than a thousandth of a percent and the sourcing shares by at most 0.4% (panels c, d), meaning that this cost works through profits rather than through prices or through the composition of sourcing. The intuition behind this is the following. When the rule tightens most firms remain compliant, as compliance is still 46 to 56% of U.S. & CA-bound revenue at the tight ends, and in order to do so they buy North American content that they would otherwise not buy. Firms are willing to pay this cost because preference is valuable, given that at observed tariffs the agreement is worth 1.4% of real income (Section 6.5). In other words, tightening the RVC threshold by 25 points costs about a fifth of what the agreement is worth, and Sections 6.2 and 6.4 show that this cost becomes larger as non-compliance becomes more expensive.

## 6.2 The Price of Non-Compliance: MFN Surcharge Sweeps

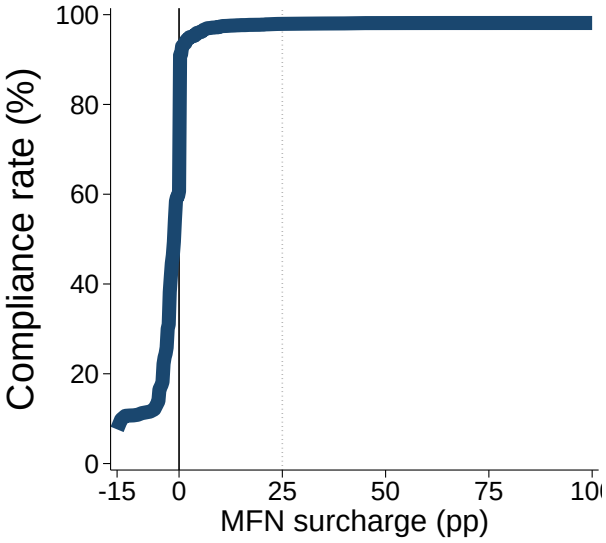
In Figure 5 I add a uniform surcharge to the MFN tariff of every product on U.S. & CA-bound shipments, sweeping it from $-15$ to $+100$ percentage points. Given that the surcharge is floored at zero, the left end of the sweep removes the tariff on every product with an MFN rate at or below 15 points, which covers essentially all of the U.S. & CA-bound revenue, whereas +25 corresponds to the design of the March 2025 order. Panel (a) shows that compliance rises steeply around the observed point and then saturates early on, going from 8% with no tariff to 61% at the observed rates, 96% at +5pp, 97% at +10pp and 98% at +25pp. This last value is the coverage ceiling, which is set by the 1.8% of revenue that has no coded rule (Section 4.2). When measured over all claims, the rate stays high even with no tariff, at 78%, and is 87% at observed rates, given that zero-MFN products count as claiming. The response is similar across the three regimes. CTC goes from 42% at observed tariffs to 99.6% at +5pp and Hybrid from 61% to 99.8%, whereas RVC goes from 80% to 95% at +5pp, 98% at +10pp and 99.5% at +25pp, showing that buying content through $\alpha^*$ pays off as soon as the tariff is a few points higher. Welfare, on the other hand, falls monotonically but slowly in panel (c), by 0.09% at +5pp, 0.14% at +25pp and 0.15% at +100pp, while the profits of the traders fall by 0.21, 0.34 and 0.37%. The reason behind this is that the marginal firms are pushed into compliance at a cost below the tariff that they avoid, and as more firms switch to preference the residual tariff bill becomes smaller, going from 0.06% of income at observed rates to 0.02% at +25pp.

This sweep also qualifies how the paper reads the 2025 episode. In the model, firms already had the capacity to comply under every type of rule well before the surcharge, as 95% of the response of compliance to a +25pp surcharge is already reached at +5pp. In the data, on the other hand, the jump from 44% to 85% between January–February and March–December 2025 (Table 3) came from the CTC and Hybrid regimes, whose rates nearly tripled, while RVC compliance only rose by six points. The model is able to match the level of RVC compliance at observed tariffs (80% against 81%), but not its small response in 2025, given that it reaches 99% at +25pp whereas the data show 84%. For CTC and Hybrid, the model gives 42 and 61% before the surcharge, against 32 and 34% in the 2024 data, and at +25pp it gives an aggregate compliance of 98%, against the 85% observed after the order. The gap at observed tariffs is the cost of claiming that the model leaves out, which is concentrated in the CTC and Hybrid regimes. The 2025 response of these two regimes, whose compliance in the data rose to 84 and 86% once the tariff was in place, shows that this cost is small when compared to a 25pp tariff.

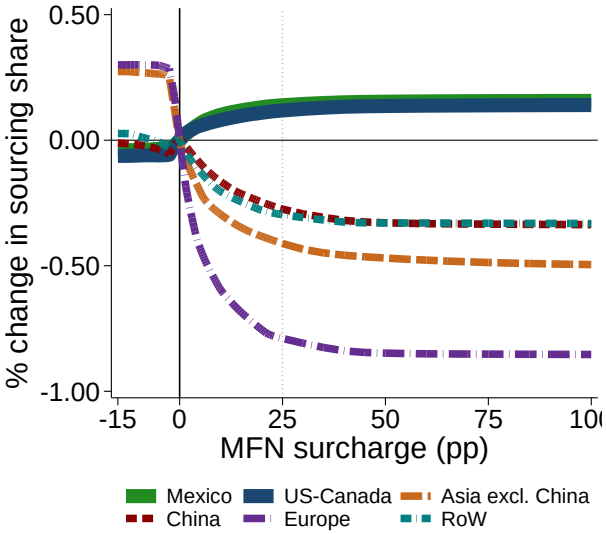
## 6.3 Tariffs on Chinese Intermediates

The next experiment is a stylized version of the tariffs that Mexico imposed in January 2026, which raised duties to 20 to 35% (up to 50%) on 1,463 tariff lines from countries without a trade agreement, China included. I model it as a uniform tariff $\tau_{\text{CH}}$ on Chinese intermediates entering Mexico, which is swept from 0 to 100% (Figure 6). Given that $\psi_s \equiv T_s(c_s\tau_s)^{-\theta}$, raising the effective input price of Chinese intermediates $w_{\text{CH}}\tau_{\text{CH}}$ by the factor $(1+\tau_{\text{CH}})$ is equivalent to scaling the sourcing shifter by $(1+\tau_{\text{CH}})^{-\theta}$, so the counterfactual enters the model as $\psi_{\text{CH}} \to \psi_{\text{CH}} \cdot (1+\tau_{\text{CH}})^{-\theta}$. The tariff revenue is rebated lump-sum to the representative Mexican consumer, assuming a balanced budget, and enters the numerator of (19) as a fiscal transfer, given that there is no tariff-revenue term there at baseline.

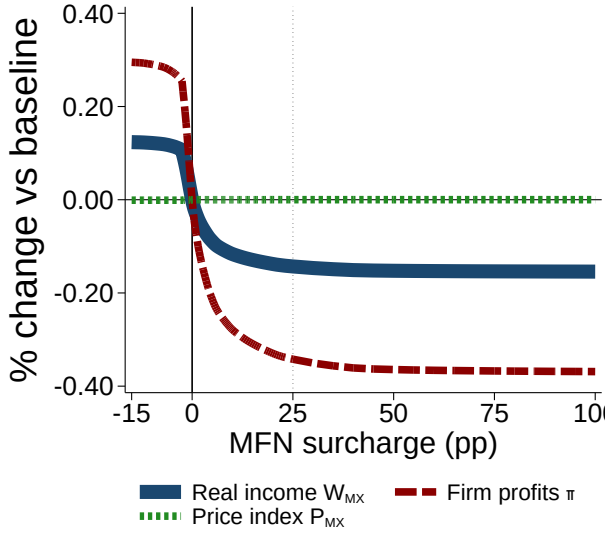
Figure 5: MFN surcharge sweep




(a) Compliance



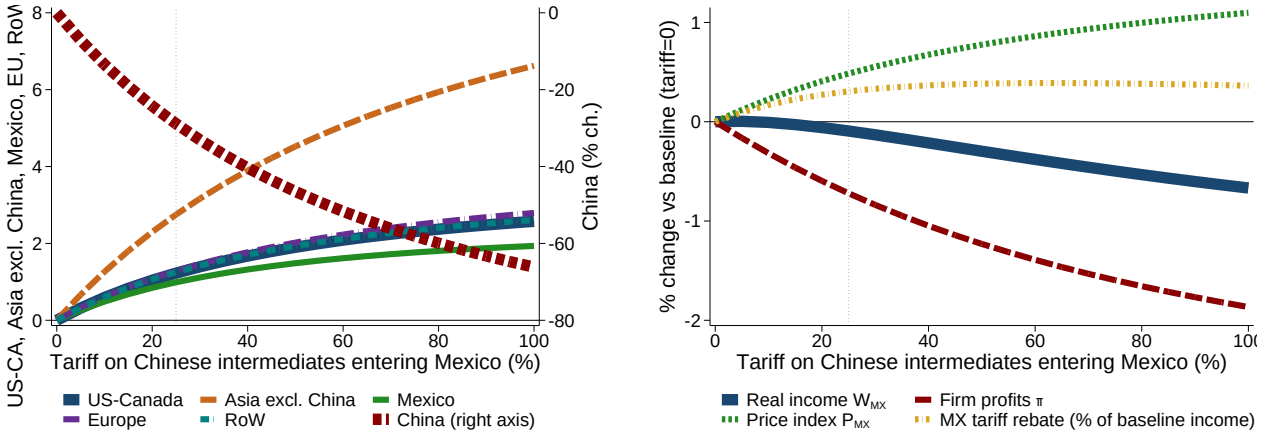

(b) Sourcing composition (shares)




(c) Welfare and components

*Notes:* 2023 vintage. Horizontal axis: surcharge in pp added to each product's own MFN rate, floored at zero, where 0 is the observed schedule (solid line) and +25 the March 2025 design (dotted line). Panel (a): compliance rate of Section 4.2 (positive-MFN lines). Panel (b): % change in each origin's share of Mexican sourcing. Panel (c): the price index moves by less than 0.001% along the sweep and its line coincides with zero. All changes are relative to the observed schedule.

Figure 6: Tariffs on Chinese intermediates entering Mexico




(a) Sourcing reallocation

(b) Welfare and components

*Notes:* 2023 vintage. Panel (a): % change in imports by origin (China on right axis), tariff-inclusive spending; dotted line at 25%.

Panel (a) shows the sourcing response. A 25% tariff on Chinese intermediates cuts Chinese sourcing by 29%, at 50% the cut goes up to 46%, and it takes a 100% tariff to reach 66%. Note that these are tariff-inclusive purchases, whereas at customs value the cuts are 43, 64 and 83%. The volume that is displaced is small in aggregate, and the source that gains the most in proportion is outside North America, as at 25% Asia excluding China rises by 2.8%, whereas Europe, the Rest of World and U.S. & CA rise by 1.3% and Mexican domestic sourcing by 1.0%. This is in line with Asia excluding China being the block whose sourcing efficiencies co-move the most with those of China (Table 12a). Meaning that the policy substitutes first toward geographies whose input cost and productivity draws are close to those of China, and that its ability to redirect trade specifically to North America is limited by this substitution pattern.

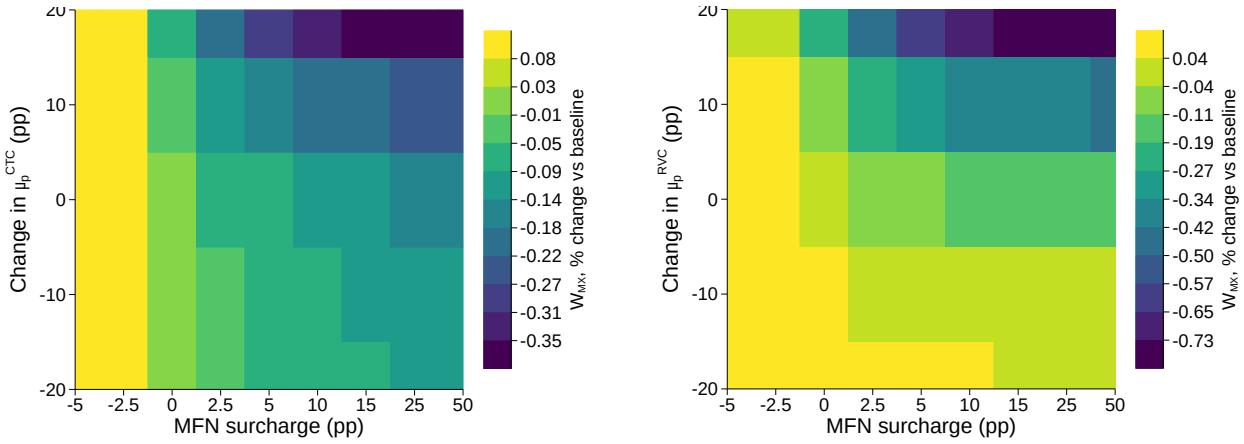
Panel (b) shows the welfare implications of the same policy. Small tariffs leave real income essentially unchanged, at +0.004% for a 5% tariff and −0.01% at 10%, given that the tariff rebate roughly offsets the fall in profits and the rise in the price index. At 25, 50 and 100%, the change in $W_M$ is −0.09, −0.30 and −0.67%, with the price index rising by 0.5, 0.8 and 1.1% and the traders' profits falling by 0.7, 1.2 and 1.9%, net of a tariff rebate worth 0.3 to 0.4% of income. Aggregate compliance rises across the sweep (from 60.7% at zero tariff to 62.2% at 25% and 64.8% at 100%), as forcing firms off Chinese inputs raises their North American content and pushes a small mass of them over the compliance threshold. When compared to the other instruments, a 25% upstream tariff costs under a tenth of a percent of real income while cutting Chinese sourcing by far more than any content-rule sweep, which leaves sourcing shares within half a percent of baseline, whereas a 25-point tightening

of the RVC threshold costs 0.3% of real income and eliminating the agreement costs 1.4% (Section 6.5). Thus, if the policy goal is to have less Chinese content in North American supply chains, upstream input tariffs work at a margin where rules of origin are essentially inert. The trade-off is that they raise sourcing from the rest of Asia proportionally more than from North America.

## 6.4 Content Rules Under a Binding Outside Option

Sections 6.1 and 6.2 vary one instrument at a time, whereas Figure 7 crosses them. Each content parameter is swept over ±20pp against a nine-point MFN surcharge grid that runs from −5pp (close to free trade) through 0 (the observed schedule) to +50pp, with +25pp being the 2025 design. Each panel shows the real-income surface that results from this.

Figure 7: Rules-of-origin strictness at different levels of the outside tariff




(a) Real income ($\alpha_p$ sweep)

(b) Real income ($\mu_p^{RVC}$ sweep)

*Notes:* 2023 vintage. Each cell corresponds to one (content parameter, MFN surcharge) pair and is shaded by the % change in real income relative to the observed schedule at baseline rules. Colour goes from dark for low to bright for high on a viridis ramp, and each panel has its own scale, shown in its legend. The surcharge steps are unevenly spaced but drawn at equal width, meaning that the horizontal axis is ordinal and not to scale.

What both panels show is that content rules matter more when not complying is expensive. At observed tariffs, tightening $\alpha_p$ or $\mu_p^{RVC}$ by 20 points costs 0.07% and 0.24% of real income, whereas at the 25-point surcharge of 2025 the same tightening costs 0.24% and 0.64%, given that firms remain compliant and buy North American content rather than paying the tariff.

## 6.5 Eliminating the Agreement

Whereas the previous exercises vary the terms of the agreement and the tariffs around it, this exercise removes the agreement altogether. Without a preferential route, every MX→U.S. & CA shipment pays the MFN tariff of its product regardless of its content, so that the compliance choice $d_{fp}$ no longer exists. The experiment is carried out on the export side only, as Mexican imports of U.S. and Canadian inputs keep their current treatment, given that the model does not carry a Mexican tariff schedule on North American inputs. Table 13 reports the outcome on the 2023 vintage along the same surcharge grid of Section 6.2.

Table 13: Eliminating the agreement

| MFN variation (pp) | −5 | Baseline | +5 | +25 | +100 |
|---|---|---|---|---|---|
| *Values,* $X^{\text{no agr.}}(s)/X^{\text{USMCA}}(s) - 1$ (%) | | | | | |
| Manufacturing profits $\Pi_M$ | −1.47 | −3.01 | −7.02 | −18.95 | −36.32 |
| Price index $P_M$ | +0.00 | +0.00 | +0.01 | +0.02 | +0.05 |
| Real income $W_M$ | −0.69 | −1.41 | −3.29 | −8.89 | −17.03 |
| *Intermediate purchases by source* $M_{MX,s}$ (%) | | | | | |
| $s$ = U.S. & CA | −2.73 | −4.79 | −9.34 | −22.84 | −42.69 |
| $s$ = MX | −1.49 | −3.14 | −7.20 | −19.17 | −36.18 |
| $s$ = AX | −1.05 | −2.63 | −7.68 | −23.21 | −46.60 |
| $s$ = CH | −1.37 | −3.39 | −8.61 | −24.32 | −48.05 |
| $s$ = EU | −1.10 | −2.41 | −6.23 | −18.58 | −36.78 |
| $s$ = RoW | −1.41 | −2.85 | −6.60 | −17.99 | −34.33 |

*Notes:* 2023 vintage (5,133 firms). Export side only, meaning that with no agreement every MX→U.S. & CA shipment pays the product's MFN rate whatever its content (zero for the 1.8% of revenue with no coded rule, Section 4.2), and compliance falls mechanically to zero. The surcharge $s$ is in pp added to each product's own MFN rate, 0 the observed 2023 schedule and +25 the March 2025 design. Both regimes are compared at a common $s$, so the tariff sweep itself is differenced out. Intermediate purchases scale with manufacturing sales at the firm choices of the executed solve.

At 2023 tariffs, removing the agreement lowers Mexican real income by 1.4%. Tariff payments to the United States go from 0.06% to 1.0% of nominal income, and the loss is spread across the whole firm block. Manufacturing sales fall by 3.4% and intermediate purchases from every source, including Mexico's own suppliers, fall by 2.4 to 4.8%, where U.S.–Canadian inputs fall the most given that the firms that lose preferential access are the ones that rely on them the most. The composition of sourcing barely changes, meaning that the loss is a scale effect on U.S. & CA-bound sales rather than a reallocation. The reason why the loss is large relative to the content-rule experiments is that a large part of the economy is exposed. U.S. & CA-bound sales are 42% of the revenue of the panel (Section 5.1), nearly all of it carries a coded rule and 70% of it a positive MFN rate, and each point of duty lowers the fob revenue of a non-compliant product by about $\sigma = 2.1$ percent. Furthermore, manufacturing profits $\Pi_M$, of traders and non-traders together, make up about half of the

income measure, with the wage bill at the pinned wage being the other half, so that about half of the proportional loss in profits passes through into real income.

Thus, what the agreement is worth depends on the tariff that it exempts firms from. The loss from eliminating it goes from 1.4% of real income at the observed schedule to 3.3% at +5pp, 8.9% at the +25pp IEEPA level and 17.0% at +100pp, and on the 2018 panel, solved under USMCA rules, the same figures are 1.4, 3.5, 9.9 and 19.2%. It is worth pointing out the asymmetry with respect to Section 6.2, where the same surcharges cost at most a sixth of a percent when the preferential route is available, given that firms respond to them by complying rather than by paying. The 2025 order, in other words, raised the value of USMCA membership to Mexico about sixfold, and the jump in compliance documented in Section 2 is the firm-level counterpart of this revaluation.

## 6.6 Two-Equilibrium Decomposition

I calibrate the model at two different vintages. Equilibrium $E_A$ is based on CE 2019 and PEME 2018 with NAFTA rules of origin and pre-trade-war sourcing conditions $\{\psi_s\}$, whereas equilibrium $E_B$ is based on CE 2024 and PEME 2023 with USMCA rules and post-trade-war $\{\psi_s\}$. Five inputs change from one equilibrium to the other, which are the relative sourcing conditions $\{\psi_s\}$, the rules of origin $(\alpha_p, \mu_p^{RVC})$, the fixed costs of sourcing and destination participation $(\kappa_s, \kappa_d)$, the foreign expenditure and price pairs $(E_d, P_d)$, and the remaining fundamentals, meaning the vintage level of the pooled imputation surface and the trader panel itself (the Fréchet shape $\theta$ is pinned at 1.79 and the sector cost shares $\gamma_s$ are frozen at their two-wave mean throughout). In order to separate these inputs, Table 14 builds a ladder of intermediate equilibria in which each of the first four inputs is flipped on its own to its later-vintage value. Equilibrium (5) flips all four of them jointly, and $E_B$ additionally re-estimates the remaining objects at the later vintage.

Letting $Y_k$ denote the value of an outcome $Y$ at equilibrium $k \in \{A, 1, 2, 3, 4, 5, B\}$, with $Y \in \{W_M,\, \Pi_M,\, P_M,\, M_{MX,s}\}$, the total change admits the exact decomposition

$$\underbrace{Y_B - Y_A}_{\text{total change}} = \underbrace{Y_1 - Y_A}_{\substack{\text{sourcing conditions} \\ (\Delta_\psi)}} + \underbrace{Y_2 - Y_A}_{\substack{\text{USMCA rules} \\ (\Delta_{RoO})}} + \underbrace{Y_3 - Y_A}_{\substack{\text{entry costs} \\ (\Delta_\kappa)}} + \underbrace{Y_4 - Y_A}_{\substack{\text{foreign demand} \\ (\Delta_D)}}$$

$$+ \underbrace{Y_5 - Y_A - \Delta_\psi - \Delta_{RoO} - \Delta_\kappa - \Delta_D}_{\substack{\text{interaction} \\ (\Delta_\times)}} + \underbrace{Y_B - Y_5}_{\substack{\text{structural residual} \\ (\Delta_S)}} .$$

Two of these components are related to the trade war. The first is $\Delta_\psi$, which captures the

Table 14: Two-equilibrium decomposition: which inputs change at each step.

| | | $\{\psi_s\}$ | $\alpha_p$, $\mu_p^{RVC}$ | $\kappa_s$, $\kappa_d$ | $E_d$, $P_d$ | Imp., panel |
|---|---|---|---|---|---|---|
| $E_A$ | Baseline | Pre-TW | NAFTA | 2018 est. | 2018 | 2018 |
| (1) | + Sourcing cond. | **Post-TW** | NAFTA | 2018 est. | 2018 | 2018 |
| (2) | + RoO | Pre-TW | **USMCA** | 2018 est. | 2018 | 2018 |
| (3) | + Entry costs | Pre-TW | NAFTA | **2023 est.** | 2018 | 2018 |
| (4) | + Foreign demand | Pre-TW | NAFTA | 2018 est. | **2023** | 2018 |
| (5) | + All four | **Post-TW** | **USMCA** | **2023 est.** | **2023** | 2018 |
| $E_B$ | Full recalibration | **Post-TW** | **USMCA** | **2023 est.** | **2023** | **2023** |

*Notes:* Bold: input set to its later-vintage value. $\theta = 1.79$ in every row. Last column: objects that are re-estimated only at $E_B$, which are the vintage level of the pooled Heckman imputation surface, the closure scalars ($w_m = A_o$, $\bar{L}$ and the domestic anchor) and the trader panel itself (5,719 firms at $E_A$ through (5), 5,133 at $E_B$). Sector cost shares $\gamma_s$ at their two-wave mean in every row. The 2018 primitives are inverted under USMCA rules, and $E_A$ re-solves that panel under NAFTA rules.

shift in relative sourcing conditions. The bundle $\psi_s \equiv T_s(c_s\tau_s)^{-\theta}$ is backed out from the observed sourcing shares, so its movement between 2018 and 2023 is plausibly dominated by the trade war and the diversion of trade that came with it. It is important to mention, however, that over this period foreign productivity, wages and freight costs also moved, and there is no way to separate them from the tariff component. The second is $\Delta_\kappa$, the entry-cost channel of the same shock. In Table 6 one can observe that the estimated source-side fixed cost falls for U.S. & CA and rises for every other block, whereas the destination-side costs rise, the most for the U.S. & CA market. The remaining components are $\Delta_{RoO}$, which is the partial effect of switching the rule parameters from NAFTA to USMCA, $\Delta_D$, which isolates the real growth of foreign expenditure (with $P_d$ held at one), and $\Delta_\times$, the non-additive interaction of the four. Lastly, $\Delta_S$ mixes several things, such as the move of the pooled imputation surface to its later-vintage level, the re-inversion of the primitives on the later panel ($\gamma_s$ is frozen across vintages and contributes nothing) and compositional turnover, given that $E_B$ is solved on the 2023 panel (5,133 firms) whereas $E_A$ and the intermediate steps are solved on the 2018 panel (5,719 firms) (Appendix C). I wish to emphasize that the identity is an exact accounting over calibrated inputs, and not a causal decomposition of shocks. The results of the six-component decomposition are shown in Table 15.

The results show that the change in the rules of origin has close to no effect on welfare, as $\Delta_{RoO}$ lowers real income by 0.03% and leaves the price index unchanged. Most of the gains come instead from foreign demand growth (+3.5%) and from the shift in sourcing conditions (+1.4%), with an interaction of +0.1%, while the entry-cost channel is small (−0.03%). In regards to the $\Delta_\psi$ component, China is the source that plays the main role. At 2018 primitives, the 2023 sourcing conditions raise imports from China by 13%, more than for any other source, and those from Europe and the Rest of World by 9 to 10%, while imports from

Table 15: Two-equilibrium decomposition: results

| Outcome | $E_A$ | Components | | | | | | $E_B$ |
|---|---|---|---|---|---|---|---|---|
| | | $\Delta_\psi$ | $\Delta_{RoO}$ | $\Delta_\kappa$ | $\Delta_D$ | $\Delta_\times$ | $\Delta_S$ | |
| *Aggregates* | | | | | | | | |
| Manufacturing profits $\Pi_M$ | 100 | +1.52 | −0.07 | −0.06 | +7.50 | +0.14 | +2.77 | 111.80 |
| Price index $P_M$ | 100 | −1.83 | +0.00 | +0.01 | −0.00 | −0.00 | −14.68 | 83.49 |
| Real income $W_M$ | 100 | +1.41 | −0.03 | −0.03 | +3.51 | +0.09 | +14.25 | 119.19 |
| *Sales by destination* | | | | | | | | |
| U.S. & CA | 100 | +2.43 | −0.13 | −0.02 | +12.02 | +0.22 | −19.90 | 94.63 |
| Mexico | 100 | +1.88 | −0.06 | −0.06 | +7.11 | +0.17 | −1.90 | 107.14 |
| Rest of World | 100 | +2.88 | +0.00 | −0.01 | +13.48 | +0.39 | −28.46 | 88.28 |
| *Intermediate purchases by source* $M_{MX,s}$ | | | | | | | | |
| U.S. & CA | 100 | −2.42 | −0.01 | −0.02 | +8.01 | −0.17 | −4.18 | 101.21 |
| Mexico | 100 | +1.49 | −0.07 | −0.03 | +7.17 | +0.14 | −3.72 | 104.98 |
| Asia excl. China | 100 | +1.89 | −0.08 | −0.01 | +8.72 | +0.15 | −18.74 | 91.92 |
| China | 100 | +12.67 | −0.03 | −0.01 | +8.82 | +1.10 | −16.90 | 105.65 |
| Europe | 100 | +8.84 | −0.44 | −0.02 | +8.32 | +0.64 | −17.88 | 99.45 |
| Rest of World | 100 | +9.73 | −0.19 | −0.04 | +7.40 | +0.72 | −7.23 | 110.39 |

*Notes:* $E_A = 100$, components in % of the $E_A$ level, $E_B$ as an index. $E_A$ and steps (1) to (5) are solved on the 2018 panel (5,719 firms), $E_B$ on the 2023 panel (5,133 firms). Peso magnitudes in constant 2023 pesos at every step (2018 census and customs values deflated with the INPC before estimation, Section 4). $\Pi_M$: profits of the traders and the non-traders. Sales by destination: traders only. $\Delta_S$ column of the price index: re-aggregation of the index over a different panel, not a price change (see text).

U.S. & CA fall by 2% and the price index by 1.8%. This sign should not be interpreted as the trade war raising Mexican welfare, given that $\{\psi_s\}$ absorbs the drift in foreign productivity, wages and freight together with the tariff channel. The structural residual on real income is large and positive (+14.3%) and is split about evenly between the price-index term and the income term. In the income term, the wage bill $w_m\bar{L}$ is 11% higher at $E_B$ because $w_m$ is measured on the 2024 census and $\bar{L}$ on the 2023 ENOE. In the price-index term, $E_B$ aggregates the CES price index over a different panel, which includes a non-trader term that doubles along with the share of the non-traders in domestic sales (18% to 30%). This means that the −14.7% move in $P_M$ under $\Delta_S$ is a re-aggregation rather than a price change, and that the level of $W_M$ cannot be compared across the two panels. Thus, the part of the ladder that one is able to interpret economically is the first four components, which are all computed on the 2018 panel. The profit row, which does not depend on the price index, shows the same ordering of the four flipped inputs, with a residual of +2.8%. Section 6.1 sweeps these same rule parameters around their USMCA baseline in isolation.

# 7 Conclusion

This paper studies how firms organize North American supply chains when tariff-free access to the U.S. market must reached with rules-of-origin compliance, and what this trade-off implies for Mexican welfare. In particular, the model allows one to quantify how much the protection that USMCA provides against U.S. tariff threats is worth to Mexico. When the tariff on non-compliant goods goes up, firms respond by complying with the product-specific content rules instead of paying it, which allows them to avoid most of the cost of the tariff. The value of holding the agreement therefore grows with the size of the threat. At the MFN tariffs in place before 202s5, eliminating USMCA would cost Mexico 1.4% of its real income, whereas at the 25-point surcharge of the 2025 order the loss would reach 8.9%, about six times as much. With the agreement in place, the same surcharge costs only 0.14%. Through the lens of the model, this is what kept the 2025 tariffs from turning into a significant economic downturn for Mexico.

The other side of this result is that, without additional tariffs, not complying with the agreement was cheap. MFN rates on Mexican manufactures are low, so many firms preferred to source without restrictions and pay them, which is consistent with preference utilization falling between 2017 and 2024 and only jumping once the 2025 surcharge raised the cost of staying outside of the agreement. For the same reason, the transition from NAFTA to USMCA rules shows to be close to welfare-neutral, given that a firm that found the new rules binding could simply pay the tariff. Foreign-demand growth and the shift in sourcing conditions toward Chinese inputs account for the interpretable part of the aggregate income change across the two vintages. Compliance responds strongly to the cost of non-compliance under every rule type, and the welfare cost of rules of origin rises with that cost.

The policy implications follow from the same logic, as how much the content rules cost depends on the tariff environment. At historical tariff levels, tightening the rules already costs real income, given that firms keep complying and buy North American content rather than lose preference. A 25-point rise in the regional value content (RVC) threshold costs 0.3% of real income, about a fifth of the value of the agreement, and a 25-point rise in the change-in-tariff-classification (CTC) input restriction costs 0.08%. At the 2025 tariff level, the cost of tighter content rules rises about threefold, above the cost of the tariff itself. Rules of origin and the outside tariff are therefore complements, in the sense that the tariff decides whether firms comply, and in turn how much the rules cost. Upstream tariffs on Chinese intermediates, on the other hand, reallocate sourcing at a welfare cost of under a tenth of a percent at a 25% rate, although they raise sourcing from the rest of Asia proportionally more than from North America. If the goal is to limit Chinese content in North American

supply chains, a tariff on Chinese inputs works at a margin where rules of origin do not.

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# A Model Derivations

This appendix collects the derivations of the closed-form expressions that are used in Section 3. Section A.1 presents the Fréchet closed forms for the intermediate cost index and the sourcing shares. Section A.2 derives the Cobb-Douglas unit cost from cost minimization. Section A.3 derives the compliant unit cost under the CTC regime, and Section A.4 derives the cheapest compliant restriction under RVC. Section A.5 describes how the firm problem is solved. Section A.6 derives the Mexican CES price index that is used in the Mexican block closure. Section A.7 decomposes manufacturing labor demand. Lastly, Section A.8 describes how the domestic-demand loop is solved.

## A.1 Fréchet Closed Forms

Conditional on the firm's sourcing set $\mathcal{S}_f$, the cost of sourcing input $j$ from country $s$ is given by

$$c_{f,s}(j) = \frac{c_s \tau_s}{z_s(j)},$$

where $z_s(j) \sim \text{Fréchet}(T_s a_{f,s}^{\theta}, \theta)$ is an idiosyncratic productivity draw, independent across countries and inputs. The firm picks the lowest-cost source for each input,

$$s^*(j) = \arg\min_{s \in \mathcal{S}_f} c_{f,s}(j).$$

Standard Fréchet algebra (Eaton and Kortum, 2002) gives the probability that country $s$ offers the minimum cost as

$$\mu_{f,s}(\mathcal{S}_f) = \int_0^1 \mathbb{1}\{s^*(j) = s\}\, dj = \frac{T_s a_{f,s}^{\theta} (c_s \tau_s)^{-\theta}}{\sum_{s' \in \mathcal{S}_f} T_{s'} a_{f,s'}^{\theta} (c_{s'} \tau_{s'})^{-\theta}}, \tag{28}$$

which is equation (7) in the main text. Integrating the minimum cost over the Fréchet distribution in turn yields the intermediate cost index

$$c_f^M(\mathcal{S}_f) = \Gamma\left(1 - \frac{1}{\theta}\right) \left( \sum_{s \in \mathcal{S}_f} T_s a_{f,s}^{\theta} (c_s \tau_s)^{-\theta} \right)^{-1/\theta}, \tag{29}$$

which corresponds to equation (6). Countries with a higher technological capability ($T_s$), higher firm-country sourcing efficiency ($a_{f,s}$), or lower effective cost ($c_s \tau_s$) account for a larger share of the firm's intermediate sourcing.

## A.2 Cobb-Douglas Unit Cost

The firm's production function combines Mexican labor $\ell$ and an intermediate composite $M$ through a constant-returns Cobb-Douglas:

$$q_f = \ell_f^{\gamma} M_f^{1-\gamma}, \qquad \gamma \in (0,1).$$

Given the Mexican wage $w_m$ and the intermediate-composite price $c_f^M(\mathcal{S}_f)$ from (6), the firm chooses $\ell$ and $M$ in order to minimize cost subject to producing $q_f$ units:

$$\min_{\ell, M} \; w_m \, \ell + c_f^M \, M \quad \text{s.t.} \quad \ell^{\gamma} M^{1-\gamma} \geq q_f.$$

The Lagrangian is $\mathcal{L} = w_m \ell + c_f^M M - \lambda(\ell^{\gamma} M^{1-\gamma} - q_f)$, with first-order conditions

$$w_m = \lambda \gamma \, \ell^{\gamma-1} M^{1-\gamma}, \qquad c_f^M = \lambda(1-\gamma) \, \ell^{\gamma} M^{-\gamma}.$$

Dividing one FOC by the other gives the Cobb-Douglas cost-share property,

$$\frac{w_m \, \ell}{c_f^M \, M} = \frac{\gamma}{1-\gamma}, \tag{30}$$

which says that the share of labor in total variable cost is equal to the production-function exponent $\gamma$. This identity is what justifies the calibration $\hat{\gamma} = wL/(wL + M_t)$ in Section 4.

Rearranging (30) gives labor as a function of the intermediate composite at the optimum,

$$\ell = \frac{\gamma}{1-\gamma} \frac{c_f^M}{w_m} M.$$

Substituting this expression into the binding production constraint $\ell^{\gamma} M^{1-\gamma} = q_f$ in turn collapses the constraint to a single equation in $M$:

$$q_f \;=\; \left( \frac{\gamma}{1-\gamma} \frac{c_f^M}{w_m} M \right)^{\gamma} M^{1-\gamma} \;=\; \left( \frac{\gamma}{1-\gamma} \right)^{\gamma} \left( \frac{c_f^M}{w_m} \right)^{\gamma} M,$$

where the second equality uses $M^{\gamma} \cdot M^{1-\gamma} = M$. Solving for $M$ and substituting back into the labor expression yields the conditional input demands

$$\ell_f^*(q_f) = q_f \left( \frac{\gamma}{1-\gamma} \frac{c_f^M}{w_m} \right)^{1-\gamma}, \qquad M_f^*(q_f) = q_f \left( \frac{1-\gamma}{\gamma} \frac{w_m}{c_f^M} \right)^{\gamma}.$$

Plugging these into the cost objective $w_m \ell_f^* + c_f^M M_f^*$ gives total cost

$$C(q_f) = q_f \, w_m^\gamma \, (c_f^M)^{1-\gamma} \left[ \left( \frac{\gamma}{1-\gamma} \right)^{1-\gamma} + \left( \frac{1-\gamma}{\gamma} \right)^{\gamma} \right] = q_f \cdot c_f.$$

The bracketed sum collapses to a constant: factoring out the common term $\gamma^{-\gamma}(1-\gamma)^{-(1-\gamma)}$, the first summand becomes $\gamma$ and the second becomes $(1-\gamma)$, so the bracket reduces to $\gamma^{-\gamma}(1-\gamma)^{-(1-\gamma)}(\gamma + (1-\gamma)) = \gamma^{-\gamma}(1-\gamma)^{-(1-\gamma)} \equiv B$. The unit cost is therefore given by

$$c_f(\mathcal{S}_f) = B \, w_m^\gamma \left[ c_f^M(\mathcal{S}_f) \right]^{1-\gamma}, \qquad B \equiv \gamma^{-\gamma}(1-\gamma)^{-(1-\gamma)}, \tag{31}$$

which corresponds to equation (9) in the main text. Note that the constant $B$ has no economic content, as it arises mechanically from the Cobb-Douglas algebra.

## A.3 CTC Compliant Unit Cost

Under CTC compliance, a measure-$\alpha_p$ subset of the firm's continuum of intermediate input varieties $j \in [0,1]$ must originate in North America. Given that input varieties are drawn iid Fréchet across countries, the law of large numbers over the continuum implies that the cost contribution of any measure-$\alpha_p$ subset depends only on its measure, and not on which specific varieties it contains. One is therefore able to treat the restricted and the unrestricted subsets as two sub-continua, each one with its own effective Fréchet scale parameter. For the restricted NA-only subset, the effective scale is $\sum_{s \in NA \cap \mathcal{S}_f} T_s a_{f,s}^\theta (c_s \tau_s)^{-\theta}$, whereas for the unrestricted subset it is $\sum_{s \in \mathcal{S}_f} T_s a_{f,s}^\theta (c_s \tau_s)^{-\theta}$. The Cobb-Douglas aggregation of the two subsets in turn gives the compliant intermediate cost index:

$$c_{f,p}^{M,CTC}(\mathcal{S}_f) = \Gamma\left(1 - \frac{1}{\theta}\right) \left( \sum_{s \in NA \cap \mathcal{S}_f} T_s a_{f,s}^\theta (c_s \tau_s)^{-\theta} \right)^{-\alpha_p/\theta} \left( \sum_{s \in \mathcal{S}_f} T_s a_{f,s}^\theta (c_s \tau_s)^{-\theta} \right)^{-(1-\alpha_p)/\theta}.$$

Substituting into the Cobb-Douglas production structure $c_f = B \, w_m^\gamma [c_f^M]^{1-\gamma}$ in turn yields the compliant unit cost:

$$\begin{aligned} c_{f,p}^{CTC}(\mathcal{S}_f) = B \, w_m^\gamma \left[ \Gamma\left(1 - \tfrac{1}{\theta}\right) \right]^{1-\gamma} \\ \times \left( \sum_{s \in NA \cap \mathcal{S}_f} T_s a_{f,s}^\theta (c_s \tau_s)^{-\theta} \right)^{-\alpha_p(1-\gamma)/\theta} \left( \sum_{s \in \mathcal{S}_f} T_s a_{f,s}^\theta (c_s \tau_s)^{-\theta} \right)^{-(1-\alpha_p)(1-\gamma)/\theta}. \end{aligned} \tag{32}$$

The intermediate-share exponents are scaled by $(1-\gamma)$ considering that only the $(1-\gamma)$ share of total cost is paid to intermediates under Cobb-Douglas production. Writing $\mu_f^{NA,M} \equiv \sum_{s \in NA \cap \mathcal{S}_f} \mu_{f,s}(\mathcal{S}_f)$ for the North American share of the unrestricted intermediate bundle, (32) collapses to the compact form used in the solver,

$$c_{f,p}^{CTC}(\mathcal{S}_f) = c_f(\mathcal{S}_f)\left(\mu_f^{NA,M}\right)^{-\alpha_p(1-\gamma)/\theta},$$

since $\sum_{s \in NA \cap \mathcal{S}_f} T_s a_{f,s}^{\theta}(c_s\tau_s)^{-\theta} = \mu_f^{NA,M} \sum_{s \in \mathcal{S}_f} T_s a_{f,s}^{\theta}(c_s\tau_s)^{-\theta}$ by (7).

## A.4 RVC Compliant Unit Cost

Under an RVC threshold the firm chooses the restricted measure itself. Restricting a measure $\alpha$ of the input continuum to North American sources leaves the unrestricted sub-continuum with NA share $\mu_f^{NA,M}$, so the NA share of the whole intermediate bundle becomes $\alpha + (1-\alpha)\mu_f^{NA,M}$ and the content share of production becomes $\gamma + (1-\gamma)\left[\alpha + (1-\alpha)\mu_f^{NA,M}\right]$. Setting this equal to $\mu_p^{RVC}$ and solving for $\alpha$ gives

$$\alpha_{f,p}^{*}(\mathcal{S}_f) = \frac{\mu_p^{RVC} - \gamma - (1-\gamma)\mu_f^{NA,M}}{(1-\gamma)\left(1-\mu_f^{NA,M}\right)} = \frac{\mu_p^{RVC} - \mu_f^{NA}(\mathcal{S}_f)}{(1-\gamma)\left(1-\mu_f^{NA,M}\right)},$$

which is clamped to $[0,1]$, so that a firm that is already above the threshold restricts nothing. Given that the restricted cost is increasing in the restricted measure, $\alpha^*$ is the cheapest compliant restriction, and the RVC compliant cost is $c_{f,p}^{RVC}(\mathcal{S}_f) = c_f(\mathcal{S}_f)\left(\mu_f^{NA,M}\right)^{-\alpha_{f,p}^{*}(1-\gamma)/\theta}$. A firm with no North American source has $\mu_f^{NA,M} = 0$ and therefore cannot comply whenever $\alpha^* > 0$.

## A.5 Solving the Firm Problem

Once the sourcing set is fixed, profits in (14) are separable across products and across destinations, meaning that I only need to enumerate sourcing sets in order to solve the firm problem. I do so in three steps.

I begin by computing, for each of the $2^{|\mathcal{S}|} - 1$ non-empty sourcing sets $\mathcal{S}_f \subseteq \mathcal{S}$, the unrestricted unit cost $c_f(\mathcal{S}_f)$ from (9) and the North American share $\mu_f^{NA}(\mathcal{S}_f)$.

I follow by noting that, given $\mathcal{S}_f$, the compliance choice $d_{fp}$ only affects the U.S. & CA revenue of product $p$, so that the firm takes the cheaper of the two routes product by product,

$$c_{f,p,\text{U.S. \& CA}}(\mathcal{S}_f) \;=\; \min\Big\{c_{f,p}(\mathcal{S}_f),\; (1+\tau_p^{MFN})^{\sigma/(\sigma-1)}\, c_f(\mathcal{S}_f)\Big\},$$

with $d_{fp} = 1$ whenever the compliant cost is weakly lower.

Lastly, given $\mathcal{S}_f$ and the compliance choices, destinations enter profits additively, meaning that the firm serves $d$ if and only if $\frac{1}{\sigma}\sum_{p\in\mathcal{P}_f} R_{f,p,d} > w_m\kappa_d$, with $R_{f,p,d}$ taken from (13). This condition takes the place of a search over the eight destination sets.

## A.6 Mexican CES Price Index

The CES price index $P_M$ in equation (18) follows from the standard demand-side dual to the destination-$d$ aggregator in (3). Starting from

$$\max_{\{q_d(\omega)\}} \left(\int_{\omega\in\Omega_d} \big(b_d(\omega)q_d(\omega)\big)^{(\sigma-1)/\sigma} d\omega\right)^{\sigma/(\sigma-1)} \quad \text{s.t.} \quad \int p_d(\omega)q_d(\omega)d\omega = E_d,$$

the first-order condition at $\omega$ gives

$$b_d(\omega)^{(\sigma-1)/\sigma}\, q_d(\omega)^{-1/\sigma} \;=\; \lambda\, p_d(\omega),$$

which can be inverted for the demand function:

$$q_d(\omega) \;=\; \lambda^{-\sigma}\, b_d(\omega)^{\sigma-1}\, p_d(\omega)^{-\sigma}.$$

Plugging this back into the budget constraint pins down the multiplier:

$$E_d \;=\; \lambda^{-\sigma}\int b_d(\omega)^{\sigma-1}\, p_d(\omega)^{1-\sigma} d\omega \;\Rightarrow\; \lambda^{-\sigma} \;=\; \frac{E_d}{\int (p_d(\omega)/b_d(\omega))^{1-\sigma} d\omega}.$$

Defining the CES price index $P_d$ implicitly through $\lambda^{-\sigma} = E_d P_d^{\sigma-1}$ in order to recover the demand expression (4), $q_d(\omega) = E_d P_d^{\sigma-1} p_d(\omega)^{-\sigma} b_d(\omega)^{\sigma-1}$, the price index satisfies

$$P_d^{1-\sigma} \;=\; \int_{\omega\in\Omega_d} \left(\frac{p_d(\omega)}{b_d(\omega)}\right)^{1-\sigma} d\omega. \tag{33}$$

In the model, varieties are firm-product pairs $\{(f,p)\}$ rather than a continuum. The set of active varieties sold at destination $d$ in equilibrium is $\{(f,p) :\, f\in\mathcal{A},\, d\in\mathcal{D}_f,\, p\in\mathcal{P}_f\}$, the appeal weight collapses to the firm-level market reach $b_{f,d}$ (shared across that firm's products at destination $d$), and the price is $p_{f,p,d}$. Replacing the integral in (33) with the corresponding sum and specializing to $d = MX$ yields equation (18):

$$\big(P_M\big)^{1-\sigma} \;=\; \int_{f\in\mathcal{A}:\, MX\in\mathcal{D}_f} \sum_{p\in\mathcal{P}_f} \left(\frac{p_{f,p,MX}}{b_{f,MX}}\right)^{1-\sigma} df.$$

Note that the $(1-\sigma)$ exponents on either side cannot be cancelled, given that the right-hand side is a sum of $(p/b)^{1-\sigma}$ terms. The equivalent rooted form $P_M = \left(\sum (p/b)^{1-\sigma}\right)^{1/(1-\sigma)}$ is algebraically identical.

## A.7 Manufacturing Labor Demand Decomposition

Total Mexican manufacturing labor compensation $w_m L_m$ has three sources across the active set $\mathcal{A}$. The first comes from the Cobb-Douglas production structure, under which a share $\gamma$ of the variable cost of each firm, $VC_f \equiv \frac{\sigma-1}{\sigma} \sum_{d\in\mathcal{D}_f} \sum_{p\in\mathcal{P}_f} R_{f,p,d}$, is paid directly to Mexican production labor, which gives the wage-bill component

$$E^{\text{prod}} = \int_{f\in\mathcal{A}} \gamma \cdot VC_f \, df.$$

The second comes from the share $(1-\gamma)\mu_{f,MX}$ of variable cost that goes to Mexican-sourced intermediates. Given that upstream Mexican suppliers are treated as making zero economic profit and as having labor as their only factor, this spending on intermediates is equivalent to the same amount of Mexican labor compensation at the upstream stage:

$$E^{\text{int}} = \int_{f\in\mathcal{A}} (1-\gamma)\, \mu_{f,MX}(\mathcal{S}_f) \cdot VC_f \, df.$$

The third comes from the sourcing fixed costs $w_m \sum_{s\in\mathcal{S}_f} \kappa_s$ and the destination entry costs $w_m \sum_{d\in\mathcal{D}_f} \kappa_d$ that each active firm pays, both of which are in units of Mexican labor by assumption, and which add

$$E^{\text{fixed}} = \int_{f\in\mathcal{A}} w_m \Big( \sum_{s\in\mathcal{S}_f} \kappa_s + \sum_{d\in\mathcal{D}_f} \kappa_d \Big) df.$$

In the executed calibration the non-trading bins are outside of the firm block (they enter through the price-index term $K$ and through their domestic sales, Section 4), so that the integral runs over the traders only. If the bins were brought back as firm rows, they would enter this same accounting with $\mathcal{S}_f = \{\text{MX}\}$, $\mathcal{D}_f \subseteq \{\text{MX}\}$ and $\mu_{f,MX} = 1$, and their variable bracket would reduce to $VC_f$. Adding up the three components gives total manufacturing labor compensation $w_m L_m$, and dividing by the pinned wage $w_m = A_o$ in turn gives the labor-demand equation (17) of the main text:

$$w_m L_m = \int_{f\in\mathcal{A}} \Big[ \big(\gamma + (1-\gamma)\mu_{f,MX}\big) VC_f + w_m \sum_{s\in\mathcal{S}_f} \kappa_s + w_m \sum_{d\in\mathcal{D}_f} \kappa_d \Big] df.$$

### A.8 Solving the Domestic-Demand Loop

Under the Cobb-Douglas upper tier (2), Mexican manufacturing expenditure moves with Mexican income,

$$E_M \;=\; E_M^0 \,\frac{I}{I^0}, \qquad I \equiv w_m \bar{L} + \Pi_M + \Pi_o, \tag{34}$$

where the superscript 0 denotes master-equilibrium values. Note that in this expression Mexican manufacturing spending moves at the baseline propensity $E_M^0/I^0$ (0.54 in 2018 and 0.61 in 2023) and not at $\chi$. The level $E_M^0$ corresponds to the baseline domestic manufacturing sales of the model firms and the non-traders (Section 4). The reason why it is anchored on these baseline sales rather than on $\chi I^0$ is that domestic sales are gross of the intermediates that firms sell to one another. Each counterfactual then iterates on $(E_M, P_M)$ in the following way. Given the pair, firms solve (14), their choices update $\Pi_M$ through (20) and $P_M$ through (18), and the loop stops once (34) and (18) hold at the firms' choices. The master equilibrium is a fixed point by construction.

## B SCIAN6–HS6 Concordance and Match Quality

The CTC input-restriction parameter $\alpha_p$ is built from the 2018 Mexican input–output (I-O) table of INEGI, which is at the six-digit SCIAN industry level, whereas the rules of origin and the model itself operate at HS6. In order to bridge the two, I rely on the NAICS6↔HS6 concordance of Pierce and Schott (2012), trade-weighted by MX→US and MX→CA export value, together with a NAICS↔SCIAN crosswalk that goes down to the deepest SCIAN level (6, 5, or 4 digits) at which INEGI reports. I then push the SCIAN I-O flows through this concordance on both the output and the input side, which gives an HS6×HS6 input–output matrix from which $\alpha_p$ is read off as the restricted-input share of each output HS6. Legal-list HS6 codes without direct I-O backing are imputed from the flow-weighted mean of their HS4 siblings, and then of their HS2 siblings, with zero as the residual fallback. Codes that belong to another HS revision take the value of the codes they link to through the UN conversion tables, and a handful of codes whose rule does not restrict any input take zero from the rule itself.

The translation shows to be successful where trade actually concentrates. Table 16 reports the match rates by treaty, both in counts and weighted by MX→U.S. & CA export value. Under USMCA, $\alpha_p$ is computed directly from the I-O for 60% of the HS6 codes but for 91–93% of trade value, given that the unmatched codes form a long tail with little trade, whereas vehicles, electronics and machinery map cleanly. The revision link accounts for 7% of codes and 7% of 2023 trade value, which are mostly codes that were renumbered at the HS2022

boundary. The direct-coding share under NAFTA is lower (49% of codes and 72% of trade value through the Annex 401 parse of Conconi et al.), with the HS2-level imputation taking in 13–21% of value. Zero fallbacks are negligible under either treaty (under 2% of codes and 0.2% of trade value), and the parameter file covers all of the manufacturing (HS2 16–97) MX→U.S. & CA export value in both periods.

Table 16: SCIAN6–HS6 match quality for the CTC input-restriction parameter $\alpha_p$

| | | | % of modeled trade value | |
|---|---|---|---|---|
| Provenance of $\alpha_p$ | HS6 codes | % of HS6 | 2018 | 2023 |
| *Panel A. USMCA* | | | | |
| Direct I-O HS6 match | 3,032 | 59.7 | 93.1 | 91.0 |
| HS4 match | 875 | 17.2 | 0.2 | 0.3 |
| HS2 match | 497 | 9.8 | 4.7 | 0.5 |
| HS revision link | 348 | 6.9 | 0.4 | 6.6 |
| Rule (no restricted inputs) | 12 | 0.2 | 0.0 | 0.0 |
| Yarn-Forward rail | 277 | 5.5 | 1.5 | 1.5 |
| Zero fallback | 36 | 0.7 | 0.2 | 0.2 |
| *Panel B. NAFTA* | | | | |
| Direct I-O HS6 match | 2,482 | 48.9 | 72.2 | 72.1 |
| HS4 match | 1,136 | 22.4 | 5.2 | 6.0 |
| HS2 match | 901 | 17.7 | 21.1 | 13.3 |
| HS revision link | 206 | 4.1 | 0.0 | 7.1 |
| Yarn-Forward rail | 258 | 5.1 | 1.3 | 1.3 |
| Zero fallback | 94 | 1.9 | 0.2 | 0.2 |

*Notes:* The 5,077 manufacturing HS6 (HS2 16–97) of the parameter file, which include codes of the HS2012, HS2017 and HS2022 revisions. Value columns weighted by 2018/2023 MX→U.S. & CA exports (UN Comtrade, Mexico as reporter).

# C CE–PEME Match Quality and Vintage Diagnostics

The trader panel comes from a direct firm-identifier join of the CE manufacturing firms onto the PEME firm×HS6 customs panel. Table 17 reports the match diagnostics produced inside the laboratory.

From the matched firms, I keep those with positive revenue, materials and wage bill in the census that trade on at least one side, and drop the exporters whose products are all Yarn-Forward, which leaves the trader panel of Table 17. Most of the larger attrition in 2018 comes from the wage-bill condition, given that before the 2021 subcontracting reform

Table 17: CE–PEME match diagnostics by vintage

| | CE 2019 × PEME 2018 | CE 2024 × PEME 2023 |
|---|---|---|
| PEME manufacturing firms | 6,854 | 6,344 |
| matched to a CE firm | 6,676 (97.4%) | 5,235 (82.5%) |
| PEME export value matched | 99.0% | 85.6% |
| PEME import value matched | 98.9% | 83.1% |
| CE manufacturing exporters ($rev_f > 0$) | 5,224 | 4,781 |
| matched to a PEME firm | 4,065 (77.8%) | 3,175 (66.4%) |
| CE foreign revenue matched | 97.0% | 86.2% |
| CE manufacturing importers ($M_f > 0$) | 3,202 | 15 |
| matched to a PEME firm | 2,360 (73.7%) | 8 |
| Trader panel (firms) | 5,719 | 5,133 |
| *PEME firm-level trade (USD)* | | |
| Exports: mean / median | \$60.4m / \$3.3m | \$84.8m / \$5.8m |
| Imports: mean / median | \$48.8m / \$3.8m | \$62.6m / \$5.3m |
| HS6 products per firm: mean / median | 27.1 / 9 | 28.3 / 10 |
| *CE foreign revenue / PEME exports, matched firms* | | |
| p25 | 0.0085 | 0.0104 |
| p50 | 0.0189 | 0.0178 |
| p75 | 0.0252 | 0.0279 |
| mean | 39.8 | 1.09 |

*Notes:* Direct firm-identifier join within the manufacturing universe. Ratio: CE foreign revenue (thousand pesos) over PEME exports (USD); exchange-rate parity ≈ 0.019 (2018), 0.018 (2023).

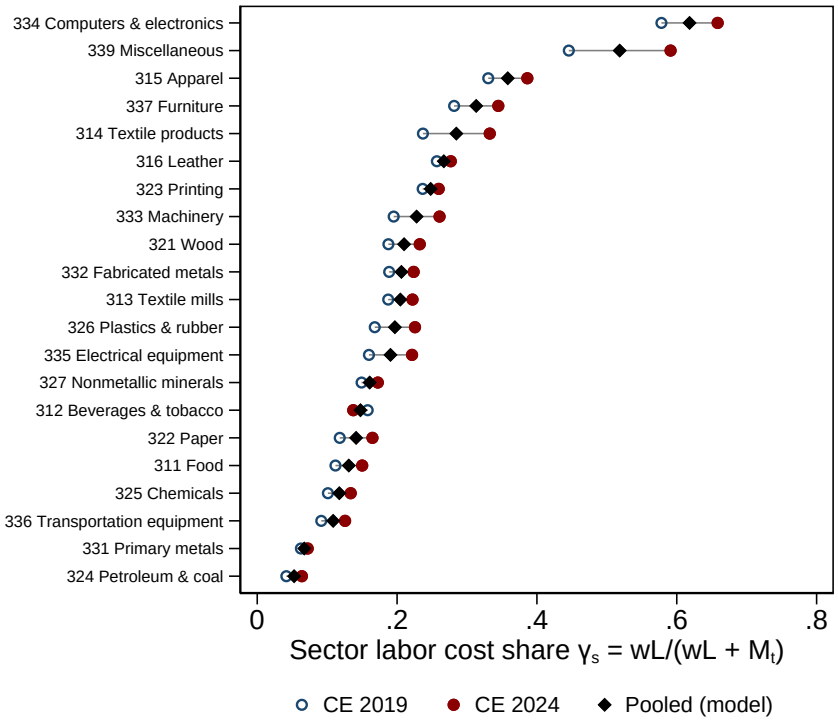
many manufacturers hired their workers through a separate staffing company and reported no payroll of their own.

The imported-materials item is effectively missing from the CE 2024 extraction (15 importers against 3,202 in 2019), meaning that the census materials of a 2024 importer already include its imports. For the non-maquila customs importers of the 2023 panel, I therefore obtain domestic materials by subtracting customs imports from census materials, with a floor at zero.

# D Discrete-Choice Fit by Block

Table 18 reports the fit of sourcing and destination participation block by block, along with the share of firms whose entire sourcing set the model is able to reproduce. Given that data and model shares coincide on the eight targeted margins by construction of the inversion, the informative columns are the "correct %" ones, which score the in/out status of each firm cell by cell. On the sourcing side, the model gets 72 to 77% of the firm-block cells right, which is above the 54 to 70% that a random assignment with the same shares would get in

Figure 8: Sector cost shares $\gamma_s$ across census vintages




*Notes:* $\gamma_s$ by NAICS-3 over all CE manufacturing firms. Open circles correspond to CE 2019, filled circles to CE 2024 and diamonds to the pooled value used in both vintages.

every block except U.S. & CA, where the high participation rate puts that benchmark at 76 to 77%. On the destination side, it gets 71 to 82% right, against 50 to 81% for the random benchmark.

# E Estimated Firm Primitives: 2018 Vintage

This appendix reports for the 2018 vintage (5,719 traders) the objects that Section 5.2 reports for 2023.

## E.1 Cross-Firm Distribution

Table 19 is the 2018 counterpart to Table 9. Sourcing efficiencies again have standard deviations of 1.5 to 1.6 log units, with Mexico having the highest mean, the AX, CH and EU blocks near a common lower mean, and RoW below them. Between the two vintages, the mean rises for every sourcing-efficiency coordinate, by 0.05 to 0.12 log units, the most for Mexico and the least for Europe. On the demand side, the mean imputed $\log b$ is 8.1 at U.S.

Table 18: Discrete-choice fit: sourcing and destination participation

| | 2018 vintage | | | 2023 vintage | | |
|---|---|---|---|---|---|---|
| Margin | Data % | Model % | Correct % | Data % | Model % | Correct % |
| *Sourcing: share of firms sourcing from* | | | | | | |
| U.S. & CA | 86.4 | 86.4 | 74.5 | 86.7 | 86.8 | 75.2 |
| Asia excl. China | 73.8 | 73.9 | 72.7 | 76.2 | 76.2 | 73.3 |
| China | 78.4 | 78.4 | 74.0 | 81.5 | 81.5 | 77.1 |
| Europe | 75.9 | 75.9 | 74.7 | 77.0 | 77.0 | 75.6 |
| Rest of World | 64.6 | 64.5 | 72.8 | 65.0 | 65.0 | 72.5 |
| *Destination: share of firms selling to* | | | | | | |
| U.S. & CA | 78.0 | 78.0 | 73.1 | 79.9 | 79.9 | 75.7 |
| Mexico | 86.1 | 86.1 | 78.4 | 89.2 | 89.2 | 81.7 |
| Rest of World | 52.7 | 52.7 | 74.3 | 55.5 | 55.6 | 71.4 |
| Exact sourcing set (% of firms) | | 37.3 | | | 38.5 | |

*Notes:* "Correct %": share of firms whose in/out status matches the data. Data and model shares coincide by construction of the inversion. MX-source participation is 100% in the model by construction ($\kappa_s^{MX} \equiv 0$), against 71% (2018) and 67% (2023) of firms with positive domestic materials in the data; omitted.

& CA, 8.9 at Mexico and 4.6 at RoW, against 8.7, 9.9 and 6.5 among the firms observed in each market.

Table 19: Estimated firm primitives: cross-firm distribution (2018 vintage)

| Coordinate | mean | sd | observed |
|---|---|---|---|
| *Sourcing efficiency* $\log a_{f,s}$ *(U.S. & CA* $\equiv 0$*)* | | | |
| Mexico | 0.259 | 1.518 | 0.58 |
| Asia excl. China | −0.656 | 1.515 | 0.70 |
| China | −0.595 | 1.507 | 0.73 |
| Europe | −0.496 | 1.496 | 0.71 |
| Rest of World | −1.006 | 1.551 | 0.61 |
| *Market reach* $\log b_{f,d}$ | | | |
| U.S. & CA | 8.567 | 2.318 | 0.78 |
| Mexico | 9.751 | 2.163 | 0.86 |
| Rest of World | 5.574 | 2.304 | 0.53 |

*Notes:* 5,719 traders. Construction as in Table 9.

## E.2 Ownership Groups

Tables 20a and 20b are the 2018 counterparts to Tables 10a and 10b, and the home-region pattern is the same. Mexican-owned firms peak in Mexico (+0.71), AX-owned firms in Asia excluding China (+1.65), Chinese-owned firms in China (+1.49) and EU-owned firms in Europe (+0.87), whereas U.S. & CA-owned firms lie below zero on every other block, and

the RoW-owned group again has no peak in its own block. On the demand side, the highest mean market reach in each market again belongs to the group from that region: Mexican-owned firms in Mexico (10.3), U.S. & CA-owned firms in the U.S. & CA market (9.7) and the remaining foreign owners in the Rest of World (6.4).

Table 20a: Estimated sourcing efficiency by ownership group (2018 vintage)

| | Ownership group | | | | | |
|---|---|---|---|---|---|---|
| | Mexico | U.S. & CA | Asia excl. China | China | Europe | Rest of World |
| Firms | 3,241 | 1,546 | 244 | 20 | 488 | 180 |
| *Mean sourcing efficiency* $\log a_{f,s}$*, by source block (U.S. & CA* $\equiv 0$*)* | | | | | | |
| Mexico | **0.71** | −0.83 | 0.85 | −0.03 | 0.40 | 0.39 |
| U.S. & CA | 0.00 | **0.00** | 0.00 | 0.00 | 0.00 | 0.00 |
| Asia excl. China | −0.85 | −0.73 | **1.65** | −0.03 | −0.39 | −0.53 |
| China | −0.64 | −0.72 | 0.05 | **1.49** | −0.29 | −0.65 |
| Europe | −0.41 | −1.14 | −0.54 | 0.04 | **0.87** | −0.19 |
| Rest of World | −0.93 | −1.48 | −1.14 | −0.66 | −0.21 | **−0.32** |

*Notes:* Construction and groups as in Table 10a. Bold: each group's home block.

Table 20b: Estimated market reach by ownership group (2018 vintage)

| | Ownership group | | |
|---|---|---|---|
| | Mexico | U.S. & CA | Other foreign |
| Firms | 3,241 | 1,546 | 932 |
| *Mean market reach* $\log b_{f,d}$*, by destination* | | | |
| Mexico | **10.29** | 8.81 | 9.44 |
| U.S. & CA | 8.06 | **9.67** | 8.51 |
| Rest of World | 5.47 | 5.27 | **6.45** |

*Notes:* Construction and groups as in Table 10b. Bold: each group's home market.

## E.3 Variation Explained by Observables

Table 21 is the 2018 counterpart to Table 11, and comes from the same regressions (27). The fully interacted observables explain at most 27% of the cross-firm variation in any coordinate (U.S. & CA demand, 0.265), and ownership is again the stronger observable for the sourcing-efficiency coordinates other than China, whereas sector is the stronger one for the three demand coordinates.

Table 21: Share of the cross-firm variation in $\log a_{f,s}$ and $\log b_{f,d}$ explained by ownership and sector (2018 vintage)

| | | $R^2$ from fixed effects for | | | |
|---|---|---|---|---|---|
| Latent | Coordinate | Own | Sector | Own + Sector | Own × Sector |
| $\log a$ | MX | 0.198 | 0.090 | 0.241 | 0.259 |
| | AX | 0.112 | 0.101 | 0.193 | 0.217 |
| | CH | 0.020 | 0.099 | 0.123 | 0.144 |
| | EU | 0.124 | 0.046 | 0.160 | 0.181 |
| | RoW | 0.055 | 0.047 | 0.097 | 0.120 |
| $\log b$ | U.S. & CA | 0.089 | 0.210 | 0.257 | 0.265 |
| | MX | 0.089 | 0.102 | 0.168 | 0.179 |
| | RoW | 0.029 | 0.098 | 0.134 | 0.142 |

*Notes:* 5,719 traders. Construction as in Table 11.

## E.4 Correlation Structure

Figure 9 reports the 2018-vintage counterpart to Tables 12a and 12b in the main text. The correlation structure shows to be stable across vintages, with the East-Asian bloc effect (the AX–CH sourcing-efficiency correlation) at 0.53 in 2018 and 0.54 in 2023, and the U.S. & CA–RoW destination correlation at 0.25 in 2018 and 0.29 in 2023. This stability supports interpreting these correlations as structural features of the trader panel and not as an artifact of a single vintage, although it is important to mention that part of it is built in, given that the two vintages share one pooled imputation surface.

Figure 9: Correlation structure of estimated firm primitives (2018 vintage)

| | MX | AX | CH | EU | RoW |
|---|---|---|---|---|---|
| MX | 1.00 | | | | |
| AX | .24 | 1.00 | | | |
| CH | .25 | .53 | 1.00 | | |
| EU | .46 | .34 | .33 | 1.00 | |
| RoW | .35 | .29 | .30 | .48 | 1.00 |

(a) Sourcing efficiency $a_{f,s}$

| | U.S. & CA | MX | RoW |
|---|---|---|---|
| U.S. & CA | 1.00 | | |
| MX | .03 | 1.00 | |
| RoW | .25 | .22 | 1.00 |

(b) Market reach $b_{f,d}$

*Notes:* 5,719 traders, 2018 vintage. Panels show the lower triangle of the correlation matrix (upper triangle omitted by symmetry). Panel (a) omits U.S. & CA ($\log a_{f,\text{U.S. \& CA}} \equiv 0$).